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Laboratory Chemistry in Natural Science

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Laboratory Chemistry in Natural Science

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Abstract

University level laboratory work is reformulated to fit into the chemistry and natural science education in the upper secondary school of Sweden. This thesis describes how this reformulation is performed both from the chemical and the didactic point of view. The resulting laboratory manual has been tested by students in the target group with positive response. Interest lay in how to present laboratory experiments far different from what they usually did and how it actually connected to their studies in natural science. How much did attitudes and self-efficacy influence the implementation of the laboratory work and how did the students collaborate? The author designed the laboratory work, observed an implementation of parts of the laboratory work through laboratory lessons in a school class and made a follow-up interview with the teacher.

Keywords: laboratory work, inorganic chemistry, reaction theory, qualitative analysis, attitudes, self-efficacy, collaboration

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Chapter 1

Introduction

For many years the interest in school science has been decreasing. Fewer students choose to study science in school and even fewer at university. Something has to be done to make school science more interesting and appealing for the students. Studies have shown that secondary school students find laboratory work the appealing factor in science teaching. Therefore making more directed and well-thought out laboratory work could be a way to stimulate students to study and learn science.

This thesis aims at describing the creation and the design of interesting chemistry laboratory work targeted at the upper secondary school. The study is based on chemical education research, but also on the author's own experiences of laboratory work at the university and in upper secondary school. At the design of the laboratory work, consequently, recollections were made about laboratory work that had left a permanent impression.

The laboratory work is focused on a laboratory section in inorganic chemistry, which is carried out at Lund University. The laboratory experiments consist of studying ion reactions and analyzing unknown samples.

The prospects were that upper secondary school students would find this laboratory work appealing, since it is playful, alive and the result is not known in advance. The laboratory work is in line with the current goals of the chemistry and natural science courses in the upper secondary school. The students are to use a mix of chemical concepts, such as solution, precipitation, pH, buffer and cushioning effects; concepts they earlier only have experienced one at the time or perhaps merely have studied in theory.

The preparation that has to be done before introducing new laboratory work within the upper secondary school might be extensive. To make the preparations easier a laboratory manual is written. This manual clearly specifies the possible goals of the laboratory work and describes the chemistry involved, both theoretically and practically. The fact that much laboratory work lacks relation to reality is often a reason that students do not feel that the laboratory work gives a lasting impression. To get a lasting impression the students should be reminded of the chemistry that takes place around them, even after having left the classroom. Therefore the laboratory manual contains background-facts and shows different areas for the compounds used in the laboratory work. The laboratory manuals can be found in Appendix A and B.

The thesis is structured as follows. Chapter 2 presents a literature study covering attitudes, learning through practical work, learning theories, and goals of the laboratory work. Chapter 3 contains the problem statement. The starting points for the creation and implementation methods used for the laboratory work are presented in Chapter 4. Chapter 5 covers the results and the analysis of the implementation of the laboratory work. Chap-

ter 6 contains a discussion and the conclusions drawn from the analysis and exercise of the laboratory work. The main text ends with a summary of consequences that this work might have on the teachers' role. The laboratory manuals for the students and the teachers are presented in Appendix A and B respectively. Appendix C presents the questionnaire and the corresponding answers of the students in the testing class.

Chapter 2

Literature exposition

2.1 Attitudes

What students think about science and how it is taught in school

An international science education project ROSE (The Relevance of Science Education) [2] shows that young people in general are positive towards science and technology. Youth of today believe science and technology enhance their life quality and make working life more interesting. But they do not like science in school because they find it hard and boring. They hesitate to work with and study science and technology and have negative opinions of scientists as persons. Students in developing countries are more interested in becoming scientists than those in the Western World. The ROSE project is a study of 25,000 young people from more than 30 countries about the concepts science and technology. It is lead by Sjøberg, the professor of Science Education at Oslo University and is developed by an international

group of scientists. A recent study by Jidesjö and Oscarsson [16] (as a part of the ROSE project) shows that Swedish students are interested in chemistry and physics, but not in the corresponding subjects taught in school.

An earlier study made as a part of the NOT-project¹ also shows that young people on the whole take a positive position towards science and technology [25], the general point of view being that science and technology is an important requirement for a prospering welfare. Students though reckon that the environment has been subjected to the technical achievements, but that only technology could save the environment [25, p. 42]. The NOT-project has also shown that the view of the teaching in science and technology among students is that it has a too weak connection to real life. Many students experience the textbooks and teaching aids, as well as the teaching methods, of being old-fashioned.

A study by Lindahl [17] implies that many pupils in secondary school still do have a positive attitude towards science² but often a more positive attitude to other subjects. Attitude is also considered being, to most students, the strongest determinant for their choice of study orientation in upper secondary school. Another great determinant (factor) is *self-efficacy*;³

¹A project funded by the Swedish government with the goal to enhance young people's interest in natural science and technology.

²Attitude in this case refers to feelings and values connected to different scientific phenomena. When attitudes are measured, the measurement is of the intensions of different actions, not the action itself.

³Perceived self-efficacy is defined as people's beliefs about their capabilities to produce designated levels of performance that exercise influence over events that affect their lives. Self-efficacy beliefs determine how people feel, think, motivate themselves and behave [6].

many students think they are good at science but not as good as in other subjects. This is in line with the theory by Bandura [7], who claims that individuals shape their lives according to what they believe they could manage to accomplish. The students' self-efficacy influence their decisions and intentions on their actions, how much effort they will put in doing it and finally how they feel during the actions. If you apply Bandura's theory, the students would choose a program where they feel they have good prospects of doing well. It is not too far-fetched to assume that studying something you consider yourself being good at and prospering in is more appealing than doing something you think is hard. According to Lindahl [17, p. 261] students believe they do not understand science as well as other ranges of subjects:

Often teachers do not make clear to the pupils what they are supposed to learn from the lab work or what is said in the textbook. When they do not understand such things they doubt their own capability and lose interest.

Bandura [7, p. 174] says that:

A strong sense of efficacy fosters a high level of motivation, academic accomplishments, and development of intrinsic interest in academic subject matter.

Bandura [7, p. 176] continues by saying that if you manage to accomplish a task you have chosen to do, the self-efficacy may enhance, and put forward that:

Educational practices should be gauged not only by the skills and knowledge they impart for present use but also by what they do to children's beliefs about their capabilities.

Self-efficacy is by Bandura, as well as by Lindahl, therefore considered being an important factor when trying to change the individual's interest in different activities. This establishes a link between self-efficacy, interest, activity and behavior. These are fundamental factors when studying attitudes.

Osborne, Driver and Simon [19, p. 30] are concerned about how school science makes students feel and put forward that:

although science interest children the experience of school science leaves many with a feeling that science is hard and inaccessible [...] such impressions of school science persist much longer than any memories of Newton's Law, the formula for sodium chloride or the characteristics of living things.

Studies by Giota [12] show that the way students experience different situations in school decides what type of motivation they will develop and with that how they will learn and develop in school over time. Giota defines the concept of motivation as the students' own goals being at school.

Echinger's study [9], on famous students looking back at their earlier teaching in science, shows that they were motivated by skilled enthusiastic and sympathetic teachers, by dynamic methods, laboratory work and discussions. But the most important factor which influenced their attitudes to science were the interaction with the teachers, especially for those students who did not have a great interest in the subject prior to the course.

In the study by Berg [8] students have been asked about what is important for them in a chemistry course. Berg concludes that they often put forward soft values like attitudes and interest, and don't make any complaints or suggestions about the chemistry involved. This implies that it is crucial to sustain the motivation of the students and boost their self-efficacy when it comes to science learning. It is easy to alienate yourself towards science and lose the grip of reality behind all theories and facts. Upper secondary students claim, according to the TIMSS⁴ study of 1998 [24], that to do well in mathematics and science you have to study hard and have great inherited talents. Furthermore, you have to be prepared to learn the book by heart to succeed in your science studies. This evidently show that studies of science in school are looked upon as products. The process that had to be prior to the product is often left aside. Sjøberg [23] discusses that science consists of both products and an everlasting process. He refers to the product of science as an art work of concepts, models and theories which today are used as tools for understanding and explaining the reality of nature. Science is not only knowing the answer, but to find answers to new questions, to broaden the knowledge and making new insights.

Jidesjö and Oscarsson [16] say that:

it seems like students apprehend the school science as being authoritarian and dead certain about things [...] this is an attitude they do not prefer.

This suggests that students are interested in sharing their view on phenomena, to start a process, but they do not get the space or the tools to do so.

⁴TIMSS=Trends in International Mathematics and Science Study.

If they cannot interact with the material they do not see the relevance of it.

Sutton [26, p. 35] discusses how textbooks present science as unquestionable facts, without focus on how the so called facts were created. He believes that science needs to be looked at through a more human perspective, stressing how the knowledge was developed. For many students:

the ideas of science become neither their own ideas nor a part of their everyday tools of thought. Those students come to regard science as just 'information' which belong to specialists, and they react to it mainly with indifference, fear and disengagement.

Sutton [26, p. 37] continues with a suggestion how to make the students participate in the scientific discussion:

One way to bring out the voice of the learner, the scientist and the teacher as personal expressions of thought is to present the lesson as a critical discussion of a scientific story (e.g., 'in this circuit, most scientists now think that...'). To work in this way places the ideas in a form in which they are open to discussion as well as to experimental test. Students are likely to explore the story and to respond with a personal voice (e.g., 'I think that what they mean is...'). Learners then have an incentive to work on their own thoughts so that a dialog can be achieved.

Berg [8] describes what he thinks is a good chemistry learning environment. That includes teachers being available for questions,

creating an atmosphere where it is acceptable to ask questions and it is accepted that everything may not be understood immediately.

Berg says that there should be:

opportunities for students to work collaboratively and exchange ideas

and stresses the possibility of connecting chemistry to other subjects and situations.

A small study made by KRC [22] (Kemilärarnas Resurscentrum) on students taking on a *naturvetenskapligt basår*⁵ shows that they are most interested in acid-base reactions and least interested in redox reactions and chemical bonding. Amongst the subjects of natural science the most interesting was biology and the least interesting was chemistry.

One of the main challenges in teaching is satisfying all students' cognitive and social needs. To do so, is to focus on what the students want to know and work more with those topics. Jidesjö and Oscarsson [16] conclude that the topics that students are interested in should be used as starting topics in teaching, to show how much science and technology they cover. Students can then work after their own abilities and at their own pace. The teacher's aim is then to provide them with the right tools to be able to plan, analyze and draw conclusions from a scientific perspective. Sutton [26, p. 31] admits that the teachers are up to a difficult task, deciding what to stipulate as 'universal' knowledge and what to be open to discussion.

⁵This is an upper secondary school year of studies in natural science, available after graduation from upper secondary school.

The problem for a teacher is to achieve an appropriate blend of definiteness and tentativeness. If you restore the human authorship and re-admit uncertainty and the possibility of argument, we can help students to gain a better idea of science-in-the-making. On the other hand, it is also a source of pride that scientists *have* managed to detach knowledge from individuals. They *have* got beyond 'So and so's opinion' and obtained agreement about facts which are understood as common to all— 'universal', 'public' or 'objective' knowledge as it is called, even by those who insist that it is consensual and more usefully thought of as 'inter subjective'.

Echinger [9] concludes that students (science and non-science majors) preferred more active and student-centered methods during their educational studies. Lindahl [17], who has made a longitudinal study about students from their sixth school year until ninth grade, observed how most students find laboratory work the most interesting part of their science studies. Some students in the sixth year in school are critical about what they actually learn during laboratory work. One student said that she often was not certain about the aims of the laboratory work done in class. She had to go home to her father and let him explain what the laboratory work aimed at. After his explanation she better understood the meaning of the laboratory work and appreciated it more than prior to the talk.

Follow up sessions with the students who appreciated laboratory work, show that the increase of interest for the science subjects are due to laboratory work in class. In the eighth year in school one girl says that she learns

2.2. LEARNING THROUGH CHALLENGES AND COLLABORATION

the most by working alone in science lessons; however, because you are able to discuss it is more fun working in groups. Eskilsson and Helldén [11] have made studies about laboratory procedures where the students are to combine laboratory work with discussions in smaller groups and between students and teacher. Many of the students found that this method of working has a positive effect on their learning.

2.2 Learning through challenges and collaboration

School science teaching has for long been influenced by Jean Piaget (1896-1980), a biologist and doctor of zoology, and his theories of the learning process. Piaget sees all mental activity as processes which give rise to creations, such as conceptions, concepts and memories [4]. The main thought is that learning takes place within the individual, where the interplay between the child and the physical world gives rise to the obtained knowledge. In the interplay the child's conceptions and memories of earlier experiences will be cognitively tested after what Piaget would call *self adjustment*. The person's aspiration for coherence makes him/her look for new explanations to the new baffling experience. Some facts could correspond with earlier knowledge, which is *assimilated* into the person's mind. In the case of assimilation, the new experience has high correlation with earlier conceptions. If the experience some how goes against what the child earlier had considered being 'true', an *accommodation* takes place. By assimilation knowledge is preserved, confirmed and enhanced, while the accommoda-

tion is progressive [10]. The will to make an adjustment is based on the child's curiosity and thirst for knowledge about the physical environment. The mind has to be challenged with new facts not quite corresponding with their earlier perceptions, to be able to evolve. All knowledge is according to Piaget built into cognitive structures⁶ which evolves through the aging of the child. Therefore the child always has some structures to build its knowledge on. His studies of cognitive structures have given rise to what today is called teaching from *a student perspective*. To be able to help a student to reach a goal we have to consider how the student elaborates and what the student already knows [4]. The student's cognitive structures put the frames around what he or she takes an interest in. What is already known is uninteresting together with what lies far beyond the existing structures. To achieve a satisfying result the teaching must lie in the outer edge of the student's cognitive structures.

Vygotsky would say that the teaching must lie in the zone of proximal development; the discrepancy between a child's actual mental age and the level he reaches in solving problems with assistance. Lev Vygotsky (1896-1934) was a Russian psychologist who considered learning foremost being a collaborative matter. You can only come to a certain level of understanding on your own. To go further you have to interplay with your environment and the people in it. In addition to this, Vygotsky means that new knowledge is born in the space between two persons in a conversation. That giving and taking is the key to achieve new knowledge. During the

⁶In Swedish: tankestrukturer; the individual's structures of thought that evolves/builds up when he or she tries to connect the new experience to earlier established knowledge and way of thinking.

1980s his thoughts became a 'vitamin injection' to the school teaching by its expansion of Piaget's theories on student learning [4]. While Piaget put the individual in focus, Vygotsky focused on the social interplay; involving family, friends, school, near society, state and culture. To be able to acquire scientific concepts and theories students have to interact with people who use these concepts while talking, explaining and having discussions [4]. From a so called Vygotsky-perspective these concepts and theories are made by humans and therefore must be presented as such.

Applying the thought of self adjustment, a practical experience could develop questions and in the long run give the individual a more complex and more consistent picture of a phenomenon. If the observations are done together with others a platform for discussion is formed and experiences can be exchanged. There could also be profits to the use of practical work as an opportunity for collaboration, because the students involved use the same discourse and perhaps easier understand each others questions and remarks. What is difficult to understand for the students is not always obvious to the teacher. Perhaps the students can draw parallels between situations and phenomena the teacher have not thought of.

2.3 Laboratory work in school science

Theory without practice is empty; practice without theory is blind **Immanuel Kant** [28, p. 204].

Laboratory work shows theoretical concepts in practice; by laboratory work you make the theories come alive. The theories are made to explain

the nature, and doing so is to give examples of nature in action. When students for instance are told about some different types of reactions, the laboratory work shows the reality behind. Learning about reactions without actually conducting them is just seeing half of the true picture. According to Hofstein [15, p. 249]:

Appropriate laboratory activities can be effective in promoting cognitive skills, meta cognitive skills, practical skills, attitude and interest towards chemistry, learning chemistry and practical work in the context of chemistry learning.

There is in other words a lot of reasons to conduct laboratory work in school environment. White [29] adds a few to the list of common used purposes, such as understanding the nature of science and also stresses on how laboratory work could influence attitudes. In his opinion the core purpose of laboratories is to assist the learning of the facts and explanations of science.

For cognitive development to work at its best there must be an interplay between theoretical studies and practical work. Practical work requires some kind of theoretical background, perhaps it is not always that evidently pronounced, nonetheless it's there. When conducting laboratory experiments in school you have to consider what aims you have.

It could for instance be to train practical skills, to observe different phenomena, or just to amuse and get the students interested in a new study section. What the teacher thinks the students are learning must not always be corresponding to what they actually learn. Students need to know why the task is done and the teacher has to recognize what was good and less

good with the task. Laboratory work that is well thought-out is easier to evaluate and change to the better. It is important to know why something is conducted in a certain way and what there is to learn from it.

During laboratory work the students will get an experience of the reactions, by feeling and seeing them in action. According to Hofstein [15, p. 250] laboratory work includes three types of practical modes:

1. problem-solving abilities,
2. skills in performing routine laboratory tasks,
3. the ability to observe and document the course of events.

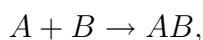
The problem-solving part in the reformulated laboratory work (Appendix A and B) includes dividing and planning the work conducted in the laboratory group. The tasks are quite basic, involving dissolving samples, adding compounds in the right order and taking good care of the reaction vessels. They have to recognize a solution, with or without precipitates, to be able to draw any conclusions.

The laboratory task is intended to be intriguing for the students. By this kind of *detective work* the thought is to show chemistry more as a process than a product (see Section 2.1). They are not expected to remember the reactions of every compound involved, but in this way they see how it can be used in a 'scientific' way. Pickering [21] wrote:

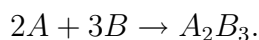
By offering a genuine, unvarnished scientific experience, a lab course can make a student into a better observer, a more careful and precise thinker, and a more deliberate problem solver. And that is what education is all about.

2.4 Chemical Theory

In a qualitative analysis samples are studied to render their contents. In the basic courses in inorganic chemistry at Lund University it is used to complement the theoretical education of chemical reaction theory and chemical equilibrium (Section 2.5). In reaction theory the university students learn how to write chemical formulae. The easiest reaction to write in formulae would be



where A and B are starting materials and AB is the product. To make it one step harder the stoichiometric amounts of A and B respectively may differ, such as



In a reaction the oxidation states of A and B can change and therefore make the stoichiometric calculations a bit more difficult. Sometimes the reacting compounds include both solid material, dissolved compounds and gases which have to be marked with (s) for solids, (aq) for solutions and (g) for gases. There are different types of chemical reactions such as protolysis, redox, complex and solubility reactions, which are further accounted for in the theory section in Appendix B.

Every chemical reaction has its chemical equilibrium. At equilibrium the reaction seems to have gone as far as possible.⁷ By changing the surroundings of the reaction, the so called *system*, the equilibrium can be dis-

⁷For a more stringent definition see "Kemisk reaktionslära" by Hägg or the basic inorganic chemistry literature "CHEMISTRY Molecules, Matter and Change" by Atkins and Jones both mentioned in the Bibliography section.

placed. A solubility equilibrium can be linked to a protolysis equilibrium. By changing the pH (within the protolysis reaction) the solubility reaction can be forced to go in the opposite direction. A solubility equilibrium can also be displaced by linking it with a complex equilibrium. At equilibrium there is a special ratio between the amounts of products and amounts of starting material. By using the products as starting material in another reaction the equilibrium is displaced. This happens in both cases – in the protolysis reaction the concentration of hydrogen ions is changed and in the complex reaction the often insoluble product is made into a soluble complex compound. More details on linked equilibria can be studied in the laboratory compendium of the KEM102-course (at Lund University) [20] or in the literature by Hägg [13].

Another way to change the system is to increase or decrease the surrounding temperature or pressure. A higher temperature can for instance make a compound more soluble. The fundamental idea is that changing the system give after-effects on the reactions within the system. That is, by changing the conditions (for a chemical reaction) you should always think of the possible results, positive as well as negative. This concept can be applied in other areas of science, such as in discussions of environmental problems and ecology.

Environmental approach

Inorganic chemistry can be linked to environmental chemistry, and in the long run also to environmental politics. Consider the possibility of combining environmental and inorganic chemistry; laboratory work gives the

basics of what reactions you could use to study the environment of a lake or a stream. Atkins and Jones [5, p. 103] mention that precipitation reactions can be used in environmental monitoring, where it may be necessary to find out how much lead or mercury there is in a sample of water. The concepts are also important in a society which depends on people having carefully prepared opinions on science development.

2.5 Laboratory history

Reactions theory in the 1960s

In inorganic chemistry the studies of reaction theory and qualitative analysis have a long history. In 1963 there was given a course at Lund University called *Oorganisk reaktionslära och kvalitativ analys för 1 och 2 betyg* [27] given at the department of inorganic analytical chemistry. The carrying out of the laboratory work and its different experiments were considered not being so difficult to implement, but they would inspire to reflections and be combined with studies of chemical equilibrium⁸ given at lectures and in the suggested course literature.

The literature in this area of chemistry was in those days concentrated to Gunnar Hägg's *Kemisk reaktionslära* [13]. This book discusses mainly chemical equilibrium in different shapes and contexts: homogeneous and heterogeneous equilibria are studied as well as protolysis, cushioning effects and solubility reactions. Hägg also gives facts on the properties of

⁸For greater insight, read for instance "Kemisk reaktionslära" by Hägg mentioned in the Bibliography section.

precipitations and how to make analyzes and separations of compounds. At the end of the book there are included studies of redox reactions, and adherent equilibrium calculations⁹.

The implementation of the experiments (in 1963) was intended to give the student knowledge of different fundamental techniques good to master in a laboratory. The student was to attain practical skills, referring to

- handling of solid, liquid as well as gaseous reagents (compounds)
- filtrations, centrifugations, and *purification*¹⁰ of precipitations
- transfer of samples from one test tube to another
- heating and evaporation
- indication of volatile compounds
- conducting melts and cleaning laboratory equipments.

During the experiments the student was to describe the reactions by formulae. Many of the reactions in the laboratory instructions concerned redox processes, and with that the student worked with balancing this type of reactions. The preparatory theory needed was given in the students' laboratory manual as well as in the course literature.

The first laboratory experiments concerned dissolution of metals, and then moved on to a more extensive study of reaction theory. In this latter laboratory section the student conducted characteristic reactions on different metal ions and composed ions. The implementation was precisely

⁹The book of reaction theory was later on supplemented by Linus Pauling's "General Chemistry" and by Gunnar Hägg's "Allmän och oorganisk kemi".

¹⁰This procedure is further described in Appendix A.

described. The student knew what the outcome was supposed to be; if there would be a generation of gas or a precipitation, even the color of it. This extensive practice worked as a foundation for the later qualitative analyses. This type of analysis was described as follows:

The task at a chemical analysis is to establish the composition of chemical compounds and mixtures. In a qualitative analysis the type of the ion kind present is determined by transforming the ions present in the sample to well defined and characteristic compounds easily observed by eyesight or sense of smell. Most identifications are made by eyesight. You observe how precipitates are formed or dissolved, color changes or generation of gas. In the latter you are of great use of smelling the compounds (NH_3 , HCN , SO_2 , H_2S).

In the qualitative analyses the student looked for both cations and anions. In the preliminary experiments in reaction theory 15 anions and 26 cations were studied. The different ions were separated through precipitation reactions. The anions were divided into three groups depending on their properties concerning

- precipitation with barium nitrate, $Ba(NO_3)_2$, in neutral environment, and
- precipitation with silver nitrate, $AgNO_3$, in acidic environment.

The ions of the first group give rise to precipitations with $Ba(NO_3)_2$, the ions in the other groups does not. On the other hand the second group

cause precipitations with $AgNO_3$. The third group does not give any precipitations in either case.

In each group the characteristic experiments were exactly described in at most six steps. The same goes for the experiments of the cations. The latter are divided into groups where the cations either cause

- chloride precipitates by the additive of hydrochloric in warm solution.
- sulfide precipitates by the additive of hydrogen sulfide, H_2S , in acidic environment. and that are a) soluble b) insoluble in sodium hydrogen sulfide, $NaHS$.
- sulfide or hydroxide precipitates by the additive of hydrogen sulfide, H_2S , in a salmiac/ammonium sulfide buffer solution.
- carbonate precipitates in a salmiac/ammonium sulfide buffer solution.

In Figure 2.1 the 26 cations are divided into groups according to the result at the above mentioned systems. In the beginning of the qualitative analysis the student was given a so called *easy* sample, which contains at most one anion and one cation kind of the earlier studied ones. When the first analysis was done and the right conclusions were drawn about the contents, a more complex sample was given to the student. This second sample contained at most four ion kinds, which could be among the 26 cations but only from six of the earlier studied anions.

The student was to go through the groups chronologically, ending up in the sixth group, if he or she were being lucky or unlucky(!). It was crucial that the student kept a precise journal on their observations, so that he/she easier could go back if anything went differently than was expected. For

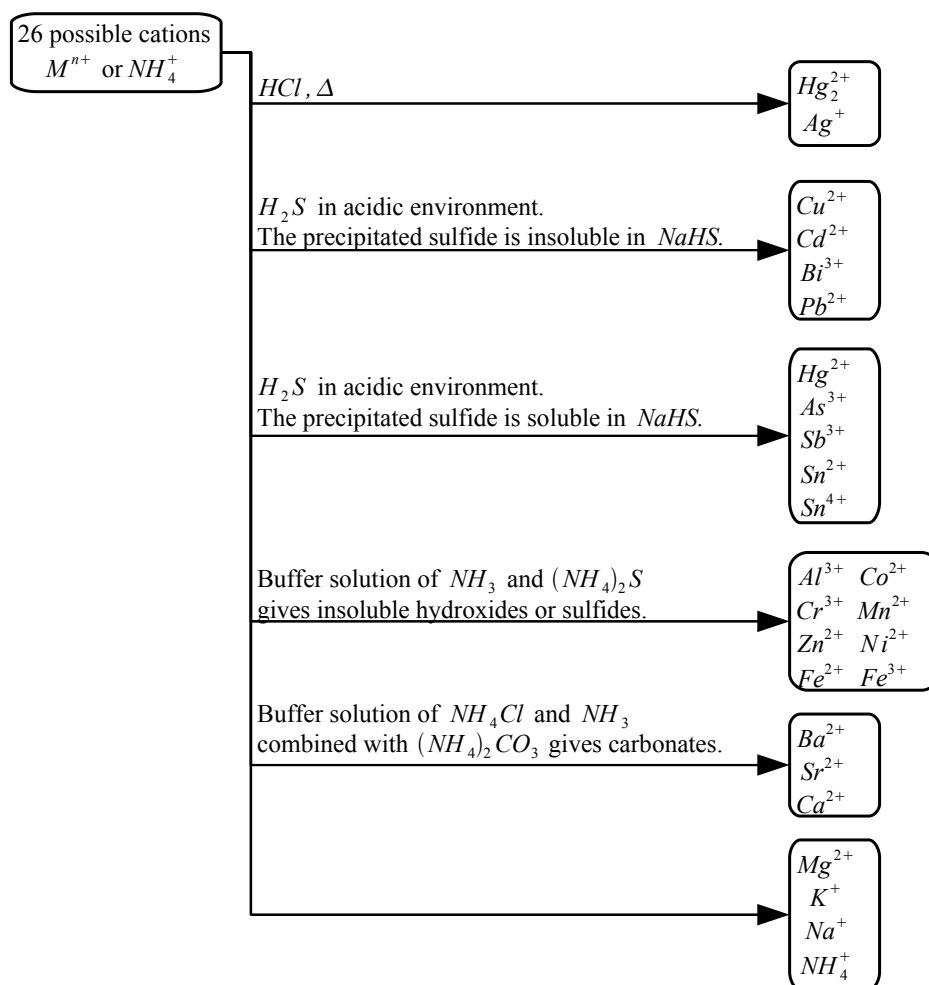


Figure 2.1: The ground structure of the separation process of cations in 1963.

instance if the student did not find the correct ion kinds he/she had to think through the last steps he/she had taken. This is an important way of conducting laboratory work, whether it is conducted on a student level or in more advanced laboratory research.

Reaction theory during the 1980s

During the 1980s the reaction theory and the qualitative analysis included less elements and was included as a part of the basic course *Kemi 1.1*. Selected reaction types were treated within the reaction theory part. Metal-dissolutions with zinc, copper and iron were studied as well as characteristic reactions with permanganate ions MnO_4^- , chromate ions CrO_4^{2-} and nitrate ions NO_3^- . Four cations NH_4^+ , Zn^{2+} , Cu^{2+} , and Ag^+ were studied starting from their reaction with sodium hydroxide and ammonia both in equivalent quantities and in surplus. The qualitative analysis was performed on 10 anions: Br^- , Cl^- , CO_3^{2-} , CrO_4^{2-} , I^- , NO_3^- , PO_4^{3-} , S^{2-} , SO_3^{2-} and SO_4^{2-} . Similar to the 1963 anion analysis (Section 2.5) the ions were divided into groups after their reaction with barium nitrate in neutral solution and with silver nitrate in acidic solution. A difference compared with the laboratory work from 1963 is that the students perform the characteristic experiments not fully knowing the grouping of the ions. After that the students had managed to group/categorize all ions, they were given a sample container one or two anions from different groups. After the analysis the students shortly accounted for their findings.

The chemistry that was used during the laboratory work included reduction, oxidation, complex bonding, solubility and precipitation. The students

were trained in handling acids, additives of as well solid as fluid compounds and to detect gases with use of a apparatus for gas examination and using the olfactory sense.

Reaction theory during the 2000s

Reaction theory and qualitative analysis have been given a large scope in the basic course in inorganic chemistry, *Kemi grundkurs 2* [20]. The student performs experiments within reaction theory that includes dissolving metals and thereafter the characteristic reactions for 16 cations and a, compared to the 1963 laboratory work, smaller number of anions: Cl^- , SO_4^{2-} , NO_3^- and CO_3^{2-} , together with NH_4^+ .

Analogous with the 1963 laboratory work the students are trained in a number of basic techniques, such as handling solid and fluid reagents, centrifugation, wash precipitations, transference of precipitations, and to point out volatile compounds. In a similar manner the students are encouraged to keep careful notes on their experiments. The students are trained on writing reaction formulae and are induced to ignore counter ions that are not involved in the chemical reactions [20].

At the implementation of laboratory experiments the students are told which cations that belong to each group and also what type of insoluble compound that is formed in each case. This is illustrated in Figure 2.2. However, the students are allowed to discover the visual results, such as color tinges/variations and any generation of gas. In the latter case the students are requested not to smell the gases. A huge difference compared to the laboratory work in 1963 is that there exists a flowchart that gives the

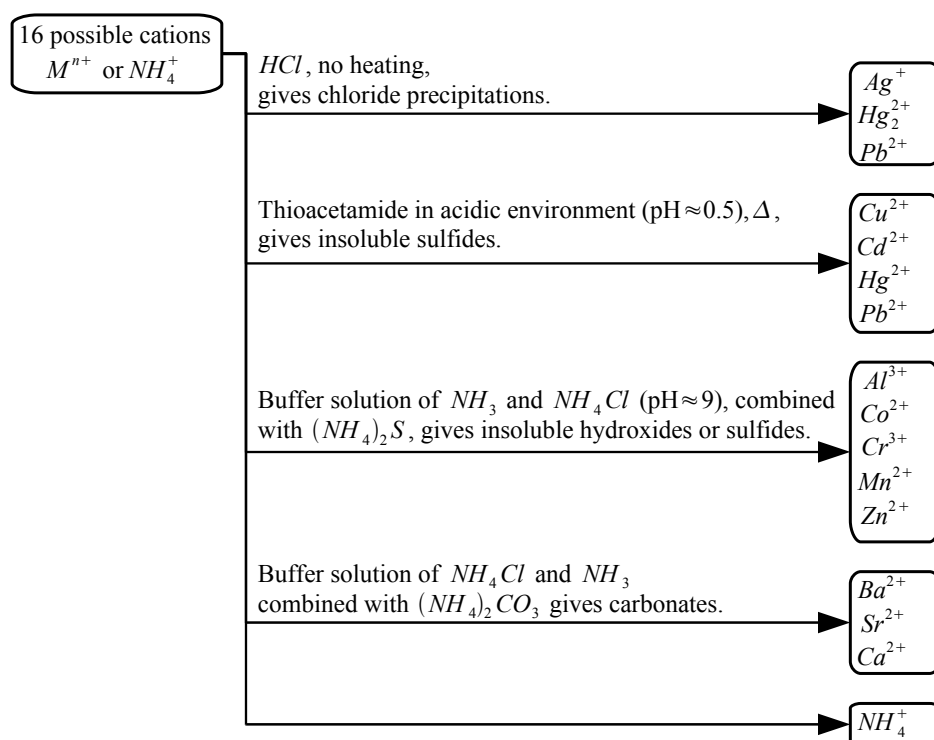


Figure 2.2: The ground structure of the separation process of cations in 2006.

students an overview during the qualitative analysis. The flowchart clearly refers to selected parts of the reaction theory that are to be used to point out a certain cation.

In the beginning the student gets to explore two sample solutions, each containing only one cation. Then a solid sample containing at most three of the cations and at most two of the designated anions. The initial step is to dissolve the solid sample; first the student tries to dissolve the sample in water and if this does not work, secondly in hydrochloric acid, or finally in nitric acid. This is done to see if it is possible to leap forward in the

flowchart.

At the end of the laboratory work the students account for the contents of their sample in front of the other students and the supervisor. Here the students get a chance to rehearse the reactions carried out during the laboratory experiments and to train on using the concepts belonging to the current section of the chemistry teaching. The time spent on the reaction theory and the quantitative analysis is five full days (10–17), including one hour for lunch.

The reactions contained in the laboratory work includes: reduction, oxidation, complex bonding, solubility and precipitation. These conceptions were also included in the 1980s laboratory work (Section 2.5) one difference however is that the number of reaction has increased and together constitute a qualitative analysis built on a successive separation of the cations in the sample through precipitation reactions, like in 1963 (Section 2.5). The chemistry behind the separation is built on the understanding of chemical equilibrium and its correlation to pH. Parts of the separation are based on the solubility of sulfides, further studied in Section B.5.

Laboratory summary from a didactic view

After having studied the laboratory work above on reaction theory and qualitative analysis, some aspects come to mind. At all times there are decisions to be made about:

- the chemical theory used,
- the risks and safety,

- the time perspective,
- the student instructions and their limitations (open/closed laboratory work), and
- the follow up work for the students.

The theory behind the reactions are more or less the same through the years: precipitation reactions, redox processes and protolysis equilibrium have always played an important role in the laboratory work. This is evident in the 1960s as well as in the 1980s and the 2000s. In the 1960s and at the millennium the qualitative analyses were more extensive handling both cation and anion analyses. In the 1980s there was only an anion analysis, but the procedure differs some from the others. The latter is a more *open* laboratory work, that is, the procedure is more exploratory for the students. The students were let to take a part of the group division of the ions. Different precipitation reagents were given, but not always which ions belonged to which group. In the laboratory procedure in 1963, the chemistry behind all the ions in each group is described in detail; what is going to happen during the experiments, including color tinges, precipitations, dissolution, generation of gas et cetera. The students are not involved in the classification of the ions.

It is easy to find similarities between the laboratory work in the 2000s and the 1960s; both involve a qualitative analysis on cations based on separation through precipitation reactions. Both seem to follow parts of the analysis schedule for instance accounted for by Moeller *et al.* [18]. There are differences in what the students are to arrive at on their own; they are

as mentioned earlier, to observe the physical results of their experiments. Only on special occasions, such as melts, the outcome chemical compound and its color is revealed on beforehand. The laboratory work in the 2000s is more open than the one in the 1960s.

There is a great difference in allocated time between the qualitative analyses. While the procedure in the 1960s seem to take at least weeks of practical work, looking at the massive manual of instructions [27], the time put in the 2000s is only about 5 days of work. By comparison of the analyses the group reagents are similar, but the amount of possible cations in each group is reduced. Amongst the removed compounds were *arsenates*, *cyanates* and mercury(I) compounds, which probably was a result of stricter rules on the risk and safety aspects. Compounds that where hard to handle through the laboratory work, such as NiS , have been removed in competition to the rest¹¹.

The theory chosen to be in the reformulated laboratory work is given in Appendix B. In the next section the laboratory lesson is accounted for in terms of goals for the upper secondary school courses Chemistry A and Science studies A.

¹¹When nickel sulfide, NiS , is precipitated it is difficult to purify (wash) it, because of its low density and low granularity. Just moving the test tubes from the centrifuge, make the precipitation "fly up" again.

2.6 Goals of this laboratory lesson in relation to courses in chemistry and natural science

In Swedish upper secondary schools chemistry is taught at the Natural Science program in two courses: Chemistry A and B. Chemistry also has a great role in the more basic/inter directional natural science courses Science studies A and B. The structure and goals of the courses can be read in Gy2000:13 established by the Swedish government. The designed university laboratory work is adapted to the courses Chemistry A [1, p. 53–57] and Science studies A [1, p. 112–115].

Purpose and goals related to Science studies A

The goals of the course Science studies A include:

Pupils should be able to make observations and carry out simple experiments, as well as be able to analyse and interpret results.

The structure of the subject implies that Science studies A is supposed to build on the students present knowledge and previous experiences. At the same time it is important to go one step further and take the teaching to another level. It is important to raise the degree of difficulty, to give the students new challenges to enhance their proximal zone of learning (earlier mentioned in Chapter 2). In the compulsory school most students have faced simpler ion-laboratory work¹². In the designed university level

¹²Laboratory work where the students for example completes a form stating between which ion-kinds precipitation took place, after having combined them on a test plate

laboratory work the students make use of this knowledge and get a new perspective. They see their knowledge used as part of a natural scientific application. It may seem not a frequent desire to separate for example chromium and silver, but performing this separation shows the students how chemistry can be used in a creative manner. Thus the laboratory work can help to create an understanding of the work process in natural science. Per definition from Skolverket¹³ this is one of the purposes of the subject Science studies A. In the text on the structure and nature of the subject it says that:

Modern society is to a very high degree based on the natural sciences and technology. For this reason, the individual needs a knowledge of natural sciences, both as an individual and as a member of society. [...] Energy, environmental and resource issues place demands on a broad knowledge of the natural sciences. The subject of Science studies is interdisciplinary, where issues concerning the natural sciences can be studied from several perspectives.

Science studies A is typically aligned towards environmental matters. When it comes to chemistry there are many topics to talk about concerning the toxicity of different compounds and their effects on the environment. If the next generation has a better understanding of the chemical reactions involved, they will have a better starting point to find ways of dealing with the environmental issues.

¹³Skolverket is an administration authority for the public educational system (in Sweden). Skolverket's part in the Swedish educational system is to e.g. state goals, inform and inspect the public sector.

"Characteristic of the subject is the mixture of theoretical studies, observations, experiments and field studies": an ambition is that the students should realize that pH has a certain role for the appearance of the environment. Questions that can be discussed in connection with the laboratory work or field studies are for example:

- What happens, from a chemical point of view, in a lake exposed to acidification?
- What happens with the metallic compounds in the water as the pH changes?
- Which types of compounds of an element are the most harmful and dangerous?

In the structure and nature of the subject, the course Science studies A is described as being "related to the pupil's study orientation". The preparatory theory for the laboratory work aims to treat the chemical contents from a perspective that the students can relate to and feel comfortable with. Science studies A is a course studied by all students no matter their direction of studies. Thus it is a difficult task to tailor a chemical background that is suited and relevant for all student groups. This includes both an adjustment to the students' interests and their earlier knowledge of science education. All students come from the 9th grade but the two earlier facts, knowledge and interest, may differ. In the student manual (see Appendix A) there are given a number of examples of subjects that can be discussed in connection with the laboratory work. One solution could be to let the students find information as a preparatory task. The idea

behind this task is that the students should find information that is related to their interests and direction of studies. This gives both the teacher and the students a chance to learn more on this particular field. The additions made by the students could be used by the teacher in the strive toward a further adaption to the different study orientations.

Purpose and goals related to Chemistry A

Many of the goals for the course Chemistry A are directly in relation to the designed laboratory work. The laboratory work could unite many of the areas that are treated in Chemistry A. Amongst the background facts associated with the laboratory work there are examples of occurrences and applications in society. In this manner the students see a connection between the chemistry they learn in class and the real life outside.

Pupils should have a familiarity with some basic elements, chemical compounds and modern materials, their properties, occurrence and processes, as well as their importance e.g. in the earth's crust or in different areas in society.

The students work with defining and balancing the chemical reactions that are included in the laboratory work. Thus, they get experience of some of the tasks in the next goal:

Pupils should be able to interpret, write and use the formula for chemical compounds and reactions, and thus reason in stoichiometric terms, as well as carry out simple calculations.

The part of the laboratory work containing chromium compounds contains an example of a reaction including an oxidation and a reduction. At

the same time an example of an application in society is given. This satisfies the goal that the students should be able to "[...] use the concept of oxidisation and reduction, and describe applications in industrial and everyday contexts."

The theoretical foundation of the laboratory work is on chemical reactions that treat solubility equilibrium and their connection to pH. The pH conception becomes a central point and the students get to see cushioning effects in a practical application. Large portions of this is included in the following goal for Chemistry A

Pupils should have knowledge of the pH concept, neutralization, strong and weak acids and bases, as well as be able to discuss equilibrium in connection with e.g. cushioning effects and be able to relate this knowledge to environmental issues amongst others.

The laboratory work treats a large portion of the chemistry included in the course Chemistry A. Thus this laboratory work can be viewed as an excellent revision of the different parts covered during the course. The students are given a lucid picture of many sections of the course. The laboratory occasion is also an opportunity for the students to use and fortify the knowledge that they have obtained and to see the concepts from a interdisciplinary perspective.

Chapter 3

Problem statement

How does the adjusted laboratory work stimulate the students' learning of chemistry? How well connected are the goals of the natural sciences courses to the actual implementation of the laboratory work?

In what ways does the laboratory work promote collaboration among the students and with the teachers during the chemistry lessons?

How does the accomplished laboratory work effect the students' self-efficacy?

Chapter 4

Methods

4.1 Theoretical and methodological starting points

When designing laboratory work based on university laboratory work it is a good idea to study earlier works of the same kind. Having studied the laboratory work of reaction theory and qualitative analyses at Lund University from the 1960s until today (see Chapter 2.5), the current laboratory work concerning reaction theory and qualitative analyses in the basic course KEM102 [20] at Lund University seemed to be a good starting point. Performing the laboratory work practically gives opportunities to discover difficulties in the implementation as well as to locate time consuming parts. This also gives possibilities to observe which reactions the students easily would be able to visually notice. The frame for the upper secondary school laboratory work is then subsequently created from the observations made during the implementation of the university laboratory work and the liter-

ature studies of other chemistry laboratory work presented in Section 2.5.

The most profitable situation would be to test the laboratory manual in a laboratory group where the students are interested in laboratory work. In this way it will be easier to see the possibilities using this special laboratory work in class. The assumption made is that there must be an initial interest in practical work combined with chemistry for laboratory work to function as a good learning environment. Especially in this case when the laboratory work is a relatively long session, taking several lessons to complete. Together with the teacher — also interested in knowing preliminary facts on the students' opinions prior to the chemistry section of the course — a questionnaire is designed. The questionnaire is designed to obtain knowledge on the students' opinions about their own knowledge in different areas of chemistry, their interest in chemistry and how they would prefer working in the classroom during chemistry lessons.

In order to study ongoing laboratory work a class belonging to the target group (upper secondary school) was selected to make observations in. Three upper secondary school classes were given the questionnaire but only one of them was especially interested in chemistry. This was also their teacher's personal opinion. There are different ways of reaching the goals of a course. Which way to choose depends on the students, each group differing from each other even though all students have studied 9 years in school. The chosen class' interest in chemistry was used to easier reach the environmental goals in Science Studies A (Section 2.6). In the other classes where the interest lay elsewhere the teacher has to work differently to reach the goals.

4.1. *THEORETICAL AND METHODOLOGICAL STARTING POINTS*

Making observations gives a chance to see and listen to the students during the implementation. Prior to this observation the possibility of using video-cameras in order to gather information was considered. However, the teacher of the class did not think that the filming would be appreciated by the students. The students had previously been asked if it was all right to take photographs during the laboratory work, in order to make recollections of the laboratory work. The students had agreed on this, but some of the students were hesitant to be photographed themselves. Hence video production was not considered as a good idea. This would probably have imposed a negative environment prior to and during the laboratory work.

The starting points for the observation are to study

- how the work is divided among the students in the groups,
- the communication and collaboration between the groups, and
- the interaction between the groups and the teacher.

The observation is followed by an interview of the teacher. The focus of this interview is on how the teacher experienced the laboratory work and the students behavior during the laboratory work. The teacher's answers to these questions are important, since the teacher can notice any differences in the students actions compared to their normal standards. The actions could relate to asking questions, seeming unsecure, motivated, bored, interested, how the groups worked together etcetera. It is also interesting to get input on the laboratory work by and large. It is chosen to make an interview since it gives the possibility of asking follow up questions. The interview will be semi-clinical, where a large number of questions is well prepared, but

the chance of asking the teacher to clarify the answers by adding further question related to the subject is allowed.

4.2 Empirical study

To be able to design the experimental work aimed at upper secondary school, there must be considerations made about the aspects mentioned in Section 2.5. A laboratory procedure is not meaningful if it is not practicable. The limitations of performing school laboratory work must be taken into account. A part of the meaningfulness must also be that the laboratory instructions stimulate student observations, hence which reactions to use must be evaluated. A laboratory course at university level could take a week or so. This possibility is not available in school. Therefore the school laboratory work must be minimized and divided into blocks of 60 to 80 minutes. Also what is allowed, available and possible to do as well as the available equipment in a university laboratory differs from the corresponding in the upper secondary school case.

Another important factor is to make the laboratory lesson exciting for the students so they get motivated to implement it and believe that they could accomplish it. To make them feel familiar with the chemistry involved in the laboratory work it has to build on their earlier experiences in the field. What earlier work the students had done was accounted for by the teacher. The intention was that a qualitative analyze and a more exploratory way of working would make the students interested of the experiments. That includes letting the students discover visual changes, such as generation of gas, precipitations, color tinges and so on. The classifi-

cation of the groups was chosen to be partly given. The students know what precipitation reagents to use to separate cations, but not the exact outcome of the reactions, for instance, they are to find out on their own which chloride precipitation that will be dissolved in hot water (see Appendix A.3). In comparison to the laboratory work in the 1960s, the one from the 2000s on the whole kept the different group precipitation reagents, but reduced the amount of cations in each group (see Figure 2.1 and 2.2). This procedure could be used also in the upper secondary school case. In this way the laboratory work could be divided into branches, where each branch could be seen as a separate laboratory section, that may be used as a separate laboratory session. Consequently the teacher could make the laboratory sections fit into the school schedule (60-80 minutes) more easily. In this way it is possible for the teacher and students to choose only to do some parts of the laboratory work, the ones best corresponding to their teaching plan. Perhaps it is not that easy to provide a full school day of only laboratory work, which would be needed. Only having two or three cations in each group makes it easier for the students to keep track of their group experiments.

The laboratory procedure was by the author's of the 1963 laboratory work [27] considered easy to implement. Together with that the separation schedule would involve much reaction theory, there will be a possibility to choose how much theory that should be handled in class. For some students the actual procedure could be challenging enough, or the insight that it actually is possible to separate ions from a sample. For others the theory behind the reactions could be an incentive to gradually understand chem-

ical concepts, such as chemical reactions, pH in connection with chemical equilibrium. The theory behind the separation and all the chemical reactions are accounted for in the teacher's manual (Appendix B). The role of the theory is to meet the students' questions. The more interested they are in knowing the theory behind the chemical actions, the more you tell them. How much to explain and discuss depends on the students. The goal is that every student would learn something more than they did before, depending on the student and his/her interest and earlier knowledge. It is a balance act for the teacher to put the theory on the right level for each student. What every student should learn though is that chemistry is dynamic, that is, changing the conditions gives after-effects. Perhaps it also would be good to think about these effects before actions are made. This is a common fact that can be applied on all human actions, not only strictly chemical. What we do to our environment has after-effects, which we now have seen in terms of global warming, climate change and animals under threat of extinction. For a certain species their conditions for living in a lake could dramatically change by the influx of water from fields, which would increase the pH of the lake.

The fact that much laboratory work lacks relation to reality is often a reason that students do not feel that the laboratory lesson gives a lasting impression. To get a lasting impression the students should be reminded of the chemistry that takes place around them, even after having left the classroom. Therefore the students' manual contains background-facts and shows different application areas for the compounds used and created in the laboratory lesson (Appendix A).

Prelaboratory work

The university level laboratory work [20] was studied and executed in its completeness. This approach gives a greater insight into the chemistry, the techniques and problem that may occur during the laboratory work. During this the different reactions and laboratory arrangements were digitally photographed. Part of these photographs are included in the final laboratory manuals. Additional photographs are available from the author on request.

Interesting metal ions according to historical features, fields of use or environmental effect were chosen to be in the school laboratory work. Facts were looked up in different chemistry books. After a discussion with Åke Oskarsson, professor of inorganic chemistry at Lund University and creator of earlier laboratory work on qualitative analyses, a reduced flowchart was created (Fig. 5.1).

Subsequently I considered the risks in using certain chemicals in school chemistry. This was done by studying texts from and in some issues contacting *Arbetsmiljöverket*¹. Chemicals that must not be used in upper secondary school teaching were taken away and substitutes were considered. To replace thioacetamide other ways to precipitate sulfides were tested: by ammonium sulphide, pure H_2S and finally the dissolution of FeS in acid. Ammonium sulphide occurred in the original lab, but only for precipitating sulfides in basic environment. The thought of using H_2S was a result of a discussion with Christina Persson, a chemistry teacher and study adviser at Lund University. During the chemical testing in the laboratory work the university laboratory instructions were followed as far as possible. If

¹The Swedish Work Environment Authority.

anything is done in a different way, it is described further in Section 5.1.

The testing of the laboratory work

On early basis contact with an upper secondary school chemistry teacher was established. This contact resulted in a discussion on the possibility of testing the designed university laboratory work in school environment. The teacher was positive to the idea of using the laboratory experiments. Unfortunately in the end there was no possibility of testing the laboratory work in her classes. Therefore another teacher, in the courses of Science studies (see Section 2.6), was contacted and the laboratory work was thought of in this use instead.

A questionnaire was handed out in the three classes of Science Studies A. In the questionnaire the students were to measure their own knowledge in different chemistry sections that are included in their course. The testing groups were to evaluate their knowledge in chemistry in average and then on concepts like the periodic system, metals and non-metals, atoms and molecules, ions and chemical bonding. There were questions about their interest in chemistry and on how they want to work during chemistry lessons. This form was created together with the teacher who was interested in these aspects in her teaching. These answers served as a foundation for the selection of testing class for the laboratory work. The results from the testing class are given in Appendix C.

The class with most interest in laboratory work was chosen to test the laboratory work, by the reasons given in Section 4.2. The teacher had already told the students about the testing when I came to visit them.

They all seemed happy about it. I visited the class on several occasions so they got accustomed to having me around.

The next step was to design students' laboratory manual. It was a must to change the language and the clarity of the instructions. Instructions were shortened and words replaced. I studied the language and layout of the chemistry laboratory instructions of the upper secondary school chemistry book *Gymnasiekemi 2* [3]. This book series by Andersson *et al.* was the main literature of the chemistry courses at the school. This book together with the corresponding literature of the Science studies course were studied to establish how they treated the chemistry related to the university laboratory work.

Dates were planned for the laboratory work so they fit in with the theory being taught by the teacher and then carried out in a couple of lessons. The teacher made in class a recollection of basic chemistry theories of Bohr's atomic model, the elements, ions, compounds and solubility. The students worked with different questions relating to this as an introduction to the chemistry section and the laboratory work. The teacher held these lessons as well as the laboratory work. I acted as an observer and helped on occasions during the laboratory work (see Part II, Section 5.3). This was done, to easier recollect the activities in the classroom. I strove for making the testing of the laboratory work into a normal school situation. It is said that observations should not be combined with activity from the observer, but I and the teacher agreed that it would feel more natural for the students if I was considered an extra teacher, helping out at the laboratory work. Anything else would have seemed odd. In this way I believe they did not

feel so exposed, as they otherwise would have been.

The students spontaneously asked me questions and sometimes said what was on their minds about the laboratory work they were implementing. There were no laboratory reports, but the students filled in answers to the questions that occurred on the laboratory instructions and some that were added. For instance they explained how they discovered the contents of different vessels in the first part of the laboratory work (Part I, Section 5.3).

4.3 Teacher interview

The observation was followed up by an interview with the teacher. Some questions were written in advance, but room was made for asking follow-up questions. These questions were the starting point for the interview:

- What made you interested in using the laboratory work within your teaching?
- In which way do you consider the laboratory work to be beneficial to your teaching and what knowledge have the students obtained from the laboratory work?
- Did you notice any difference in the manner the students acted during the laboratory work in comparison to how they usually do during these lessons?
- How well do you consider the laboratory work to fit within the teaching framework of your course?

- Have you considered to continue using the laboratory work—complete or in parts—in your teaching and in what manner?
- Do you have any suggestion on changes to make the laboratory work better?

The results of the interview are found in Section 5.8.

Chapter 5

Results

The results and analyses are presented as follows: first the laboratory actions made during the creation of the laboratory work are explained, then follows the results from the questionnaire (Section 4.1) and the class observations from the carrying out of the laboratory work. Subsequently analyses are made from the observation and the written answers from the student laboratory manual in connection with the possible goals (Section 2.6) of the laboratory work. Thereafter follows a summary of the interview taken with the teacher of the class.

5.1 Chemical aspects when formulating the laboratory work

It took about two weeks trying out and studying the university laboratory instructions. One of the experiments given in these instructions was to precipitate lead chloride, $PbCl_2$, after having it heated on a water-bath. At

the first try, this did not work out. The reason found was that it had prior to the cooling been heated too long. The heating was done to dissolve the solid $PbCl_2$. The reaction tube was left aside while handling the photos taken during the prior experimental procedures, approximately 15 minutes. When the experiment was done without other distractions, the compound did precipitate after cooling. This is an experience though to remember and think of as a possible difficulty in the student implementation.

To shorten the time of the laboratory experiments only a number of metal ions from the university laboratory instructions were chosen to participate in the new laboratory work. These metal ions were Ag^+ , Pb^{2+} , Cu^{2+} , Mn^{2+} and Cr^{3+} . The facts on these ions are included in the manual of the laboratory work. Fig. 5.1 shows the flow chart made from these metal ions.

Only chemicals belonging to the B-list are allowed to be used in school environment. In the university laboratory instructions thioacetamide is used to precipitate sulfides in acidic environment. This chemical is A-listed, because it is carcinogenic. The first attempt was to use ammonium sulphide instead. The laboratory instructions were executed according to the flow chart using ammonium sulphide. This took about 45–60 minutes.

The side effects on using ammonium sulphide is the need of having it newly-made. A once opened flask is good to use for about a week if held in a fridge. Trying to use it after this time gives rise to side reactions including greasy polysulphur-compounds. I imagined that there could a problem sustaining the solution acidic when using ammonium sulphide to precipitate CuS and PbS . To test this solutions of $Mn(NO_3)_2$ respectively

$Co(NO_3)_2$ were made highly acidic with HCl (pH 1) and ammonium sulphide was added drop-wise. By the first drops there were precipitations. The precipitations came as shadows. The pH showed it to be acidic.

Solutions of CuS and PbS were cautiously made acidic by adding 0.1 M HCl drop wise until pH reached 1. After adding 5 drops of ammonium sulphide, pH was measured and as many drops were added so that the pH again reached 1. In both cases (CuS respectively PbS) 25 drops of acid were needed. These amounts were further checked by making combinations of Cu^{2+} together with Mn^{2+} respectively Co^{2+} . The same was done with Pb^{2+} together with Mn^{2+} respectively Co^{2+} . To sustain the volume of the system, half the amount (7 drops) of nitrate solutions were used in the two cation species. In each case the pH was sustained at pH 1-2 after the adding of ammonium sulphide (5 drops). The precipitations were centrifugated and the solution was made basic by the adding of 2 drops (5M) ammoniac which gave the other expected precipitate. The pH of the solutions were now around 8.

It seemed a bit tricky playing with the pH, so I looked for other solutions. The ammonium sulphide give rise to H_2S which then gives S^{2-} to the metal sulfides. By just skipping the ammonium ions the pH would not be a problem anymore or would it? This had to be investigated.

H_2S gas was bubbled into 1 ml samples (15 drops) of 0.1 M nitrate solutions. That gave rise to the expected sulfides in about a few seconds up to a minute or so to get full precipitation. One side effect was the contamination of the air, even though it all as executed in a fume-cupboard. The smell and the fact that I felt dizzy were signs on that. H_2S only smells

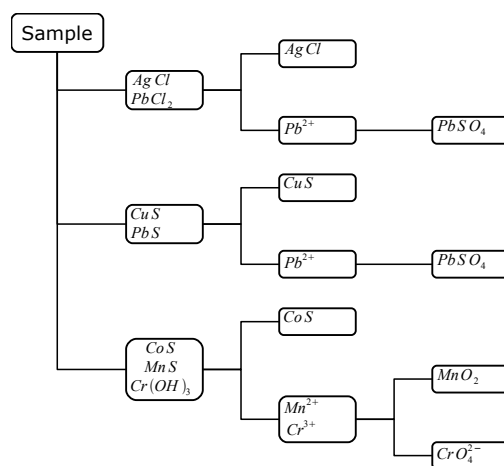


Figure 5.1: The flowchart of the separation procedure.

at first, later your smelling system gets numb and you get a dizzy feeling.

The next thing to try was the dissolution of FeS in HCl . The chemicals were put in a gas flask and the generated gas was lead to the test tubes containing nitrate solutions. For this a *gas flask* was used where 5M HCl and a few chunks of FeS . The gas (H_2S) was led to the other tube where the metal nitrate solution was put. This is illustrated in Fig. 5.2.

5.2 The questionnaire

In the beginning of the chemistry section in Science Studies A, the students were given a questionnaire (Appendix C). On a scale—weak, good, very good, excellent—most students regarded their knowledge in chemistry of being weak or good (76.8%, Table C.1). After the chemistry section, when they were asked again, they believed their abilities had evolved and answered good or very good (88.9%, Table C.2). The results show that they did not like to study chemistry by just answering questions in their



Figure 5.2: The arrangement for the FeS reaction. The outgoing pipe in the upper right corner is connected to a tubber finger-stall.

books (61.5%, Table C.1), some would think it was ok (30.8%), but only few wanted to do it very much (7.7%). Most students wanted to do laboratory work often (46.2%), some of them in every lesson (30.8%). Everyone wanted to do laboratory work to some extent.

5.3 Observations from the testing of the laboratory work

Some parts of the laboratory work were chosen to be in the students' natural science course after discussion with the teacher. The first part being Laboratory section I (see Appendix A.3 and B.6), where the students han-

dle chloride precipitations. Thereafter one kind of analysis was made of this Laboratory section, where the student groups get three test tubes; two with one cation kind each, one with the two cation kinds (see Appendix B.4). Subsequently Laboratory section III was implemented (Part II), concerning chromium and cobalt compounds (see Appendix A.5 and B.6). Below follows a summary from the activities taking place during the laboratory work. In the observations focus lay on the interactions between students and teacher–student interactions, and how the students reasoned in their groups on the chemistry experiments and instructions.

Part I The students worked in teams of two or three persons. During the first laboratory section meeting they performed the characteristic experiments in order to learn how to recognize silver(I)- and lead(II)-ions. The teams observed that the time before precipitation takes place is different and also that granularity differs between the different compounds. One of the students explained:

For lead chloride it looks like it is snowing!

and continued the statement by explaining that it takes longer time before something happens with the lead chloride compared to the silver chloride. One of the groups considered the silver chloride as being slightly pink colored. In total the carrying out of these tasks took about half an hour.

The students seemed to be positively inclined to test the laboratory session. One of the students told me that it was fun to do something a bit more difficult than they usually did. The teacher had prior to the laboratory work told the students that I was coming to visit them and that

they were going to perform laboratory work that usually takes place at Lund University. The students felt chosen and entrusted with something above the normal. After the first part of the laboratory testing it was clear that they felt proud and elated at having succeeded in completing the laboratory work.

During the next lesson each of the groups were given three test tubes: one containing silver(I)-ions, one with lead(II)-ions and one containing both silver(I)- and lead(II)-ions. The task given to the students was to find the contents of the test tubes. Most of the groups came up with suggestions on different procedures on how to find the answers. The procedures were mainly based on the observations the students made during the first meeting. One student said:

In two of the test tubes precipitation took place directly.

Then the student continued this statement on commenting one of the test tubes (the test tube containing both ion kinds) in more detail by saying:

At first there was a precipitation, then after a while more precipitation took place... the latter looked more as it were snowing... and had larger flakes... thus it looked as the lead chloride.

In order to test their hypothesis the student continued by following the laboratory manual to make sure that there assumptions were correct. All of the groups succeeded in following the manual and to establish the contents of the test tubes. The carrying out of this part required approximately half an hour.

Some of the students participating in the second meeting had not been to the first meeting. In order to facilitate the participation for these students the teacher suggested that the students that attended the first meeting should help the new ones. In one of the groups the student having attended the first meeting accepted responsibility and showed two other students in the group how to *wash* precipitations. Since they were to *wash* three precipitations simultaneously all members of the group could participate in this element of the laboratory lesson. The student also explained how to differentiate between lead chloride and silver chloride by heating the solutions for the other two students. One student, who usually is not in the center of attention at the lectures, also took the responsibility of leading the group. A responsibility that the student carried out successfully and seemed elated at.

After the two meetings in the first part of the laboratory testing all students were given time to write down their observations and to answer the questions that were given in the laboratory manual. Here the students described the colors of the precipitations, how the precipitation took place in the different cases as well as writing a short description on how they conclude what was in the different test tubes. During the time the teacher walked around in the classroom and participated in the students' discussions.

The next lesson worked as a feedback session where the teacher and the students talked about what they had learned about ions, salts and their solubilities.

Part II At the beginning of the second part of the laboratory testing the students performed the characteristic experiments in order to learn how to recognize cobalt(II) and chromium(III). All chemicals and all equipment were put on a separate table and the student had to provide themselves with what they needed during the different steps of the laboratory work. One of the neighboring rooms, equipped with a fume cupboard, was used as a preparation room to produce hydrogen sulfide. Part of the chemicals in the laboratory work, such as hydrogen peroxide and nitric acid, was located in a separate fume cupboard. The students performed the initial steps in the laboratory manual. When they arrived at the point where they were to precipitate cobalt sulfide, they went to me for assistance. From the gas bottle containing a reaction mixture of iron sulfide and hydrogen chloride, I *bubbled* hydrogen sulfide into their test tube. The students observed that a black precipitation was formed during this process.

In order to avoid spreading hydrogen sulfide in the classroom, the students had been requested to cork their test tubes prior to moving them towards the centrifuge located in a fume cupboard in the middle of the room. At one occasion one of the students opened the test tube. As a result it started to smell terrible. The students located near the place of the scenario told the student that opened the test tube to close it again, and so he did. Close thereafter three windows were opened by students and the odor quickly disappeared. This setback did not prevent the students from going on with the laboratory work, being as motivated as prior to the accident.

Subsequently the groups came to me to get an additive of hydrogen

peroxide into their test tube containing chromium(III). They all noticed that the solution turned yellowish. When boiling the solution part of the groups reaction mixtures turned more towards orange/brown than yellow. After adding lead nitrate a yellow precipitation was formed in some of the groups solutions while an orange precipitation was formed in the others. This got the students slightly confused. This matter is further discussed in Section 5.4.

The time allocated to this part of the laboratory work was an 80 minute lesson. However due to the schools ventilation system was turned off—and thus the fume cupboards could not be utilized until the ventilation worked normally again—during the beginning of the lesson, the students only had 60 minutes to complete the laboratory work. The students still managed to finish all instructions of the laboratory work within the reduced time frame. At the end of the lesson test tubes from different groups were selected. The following lesson these test tubes were used to illustrate the different experiments.

At the next meeting the laboratory work and the reactions involved were discussed in class. When the precipitations from the laboratory lesson were displayed one of the students, who had not been participating in the laboratory lesson, walked up to them for a closer look. The student lifted one of the tubes and looked at it carefully, then the precipitation was put back into place and the student returned to the others. The students and the teacher together reviewed and orally reconstructed the laboratory work and its happenings. Overheads showing parts of the laboratory manual and photographs were displayed and replaced throughout the review

process. The different experiments were discussed at the same time as the corresponding test tubes were displayed. In some cases the teacher walked around and showed the photographs taken during the laboratory lesson (at the time when all the precipitation were new).

The teacher explained that the purpose of the reaction was to show that it is possible to create and dissolve precipitations by changing the pH of the solution. Thereafter the teacher explained how the pH-value is connected to the concentration of hydrogen ions. This issue had been mentioned prior to the laboratory work, but not to this extent. The teacher regarded the theory (of pH and its connection to precipitation reactions) to be more understandable now when the students had seen it in practice. It was also assumed that it is more interesting finding the essence of the laboratory work together with the students, in terms: "What have we learned?" more than "We do this because...".

The teacher displayed the test tube with the yellow sample and asked the students if it looked the same as just after the reductive agent had been added. The class agreed that is previously looked more like a cola soft drink. The teacher asked the students:

What did you do to make the contents yellow?

One student answered

We boiled!

Then the teacher continued by asking

What did you boil away?

to this question the same student answered

Hydrogen peroxide.

The teacher continued the review by explaining that the acidity has a large influence on into which compound $Cr(III)$ will be oxidized, that there are different reactions taking place depending on pH. The teacher also commented on two of the groups who got the reaction giving rise to an orange compound, where the chromium had turned into dichromate. The chemistry behind is further mentioned in Appendix B.6.

After the feedback session the students finished writing their reports, that is, they continued answering the questions from the laboratory manual and handed them to the teacher. Thereafter the teacher made a transition to the energy section which was the next subject to be discussed.

5.4 Collaboration during the laboratory work

In the analyses of the laboratory work, the collaboration between the students within and beyond the lab groups were studied and also how they reasoned in their groups on the chemistry experiments and instructions.

During the second part of the laboratory testing (Section 5.3) there were clear differences in how the groups divided the work among their members. Four group types could be distinguished, henceforth denoted Group I, II, III, and IV. Group I consisted of two students that carried out all instructions thoroughly, and seemed to be committed to succeed with the laboratory work. Student M held the initiative and was the driving force, while student D was more of a sidekick och took front the material and equipment needed

and let M complete the chemical instructions. Clearly student M was the leader of the group. When the teacher walked around and checked on them, they did not want any help. They assured the teacher that they knew what they were doing.

Group II consisted of three students who established distinct roles after a short while. One of the students performed the chemical instructions; one wrote down observations, after discussions with the other members of the group; and the third students read the laboratory manual and checked what was to be done.

Group III consisted of two students, who performed all of the different parts of the laboratory instructions—practical as well as written—together. They communicated with each other and asked for the others thoughts and strengthened each other. Asking each other questions like: -I think it looks like [...] what do you think? The students in this group were, according to their teacher, used to working together during laboratory work and other tasks.

Group IV consisted of two or three students, where one of the students performs the laboratory tasks and the other has a more passive role. This group was happy and positive, but did not seem to have a great confidence in their own abilities. They felt insecure on their work and often asked the teacher if they had succeeded in the last step. On some occasions they asked the students in other groups what results they had got after a certain part of the experiments. The questions asked was of the type "Should it become green?". They did not want to show what they had got until they got their answer. Then they gladly said "We got that, too...", when that

was the case. They succeeded in getting through the laboratory work, and carrying out all parts of it. The time spent on this was larger than for the other groups and the teacher got to help them and explain and show some procedures and act as backup when the students felt insecure on what to do.

Comparisons of results between the groups happened on and off during the laboratory work. In this way the laboratory work created a situation where the students felt the need of discussing their results with other groups. If they had believed the tasks were easy, this would probably not have happened, at least not to this extent. The teacher commented in the interview (Section 5.8) that she observed the groups discussing with each other, which she was happy about. It was easy for a group to start a discussion with others because they had one group beside them and another group just behind them. When something happened in the nearby fume cupboard or when they took something out more than themselves in the group were able to see the results, and comment if they wanted. When the teacher walked around and talked to the groups, she was at the same time near the other groups and discussions between the groups could easily arise.

5.5 The students' written notes (from the laboratory work)

The students handed in all work they did during the course, for example answers to questions from the course book or working papers. The students handed in their notes on from the laboratory work, both the analysis discus-

sion from Part I and their observations from Part II (see Section 5.3). The students were also to write chemical formulae, which they did either during or after the laboratory work in class. Recollections from the discussions made in connection with this task is also added to the analysis.

In the student manual instructions questions were added to the experiments. The students gave their notes on the questions in Laboratory section I (Appendix A.3) and Laboratory section III (Appendix A.5). In Laboratory section III there were lines marked for observations. Below follows some examples of the answers from Laboratory section I.

Referring to what had happened after the adding of hydrochloric acid to the silver(I) nitrate the student answers were for instance

It became white.

White precipitation was formed.

After the precipitation of lead(II) chloride the students wrote

You see that it is snowing (red. in the test tube).

It looks like snow flakes that are sinking.

In the end when the students accounted for the results after having heated the chloride precipitations they wrote

The lead chloride dissolved but not the silver chloride.

When you heat them up the lead chloride dissolves while the silver chloride remains (red. in the state it was prior to the heating).

After the characteristic experiments the students received three test tubes and their task was to analyze the contents. All the test tubes were on before hand numbered. The group receiving number 4 (silver(I) and lead(II)), 5 (lead(II)) and 6 (silver(I)) reasoned (in writing) as follows:

4 = Silver/lead, that you see when you boil it (red. the test tube) and cool it down again, because then there are precipitations.

5 = When we heated it (red. the test tube) the solution was dissolved (red. the contents become a solution), and that was the earlier case with the lead chloride.

6 = When we had heated and cooled (red. the test tube) there were no new precipitations, therefore it is silver chloride.

The students also added that they had observed that the precipitations of the two chlorides did not look quite the same and that they had a hunch by the appearance of the precipitations that they found in the test tubes. One group relied on these visual observations and wrote:

Considering the 'snow' in test tubes 1 and 3 we directly noticed that the contents of the tubes could be

1. Lead chloride
2. Silver chloride
3. Both

The same group made the comments on the precipitations in one of their test tubes taking different amount of time taking place (Part I Section 5.3).

Considering this there should, together with the 'snow' reflection, therefore actually be two different precipitations in the test tube.

The students used their earlier observations to analyze the contents of their test tubes. They relied on their observations on the properties of the chlorides when they were heated, their appearance and the fact that reactions could take different amount of time before results can be seen. The students used concepts like 'solution', 'precipitation', 'reaction' and chemical names (silver chloride and lead chloride). This could be related to the goal referring to the students

[...] should be able to make observations and carry out simple experiments, as well as be able to analyse and interpret results

which was one of the goals of the laboratory work [1]. In connection with Laboratory section I the students were to write chemical formulae for the chlorides: $AgCl$, $PbCl_2$. Most of them got it right. Some students were insecure about which came first the 'Cl' or the 'Pb', how many 'Cl' there should be and where the numbers should be put. After a discussion on the standard way of writing formulae between the group and the teacher, this was sorted out.

In Laboratory section III (Appendix A.5) the students were to observe differences in the properties of cobalt(II) and chromium(III). The students gave answers about the color of the precipitations, their names and chemical formulae, wrote chemical reactions and noticed the generation of gas.

One group used ammoniac instead of the pH 9 buffer, but got the same results as the others:

A drop of ammoniac and there becomes a pail green precipitation

which was corresponding to the answers of the other groups. The students wrote observations about the formation of cobalt sulfide, that it smelled like "crap" or "rotten eggs". These students belonged to the group where one student opened their test tube outside the fume-cupboard (Section 5.3). After the adding of the hydrogen peroxide the students noticed that the contents of the test tube became "brownish" and that there was a generation of gas. No one wrote any assumptions on what gas it was.

The students had not handled reduction and oxidation reactions in formulae before, so the task of answering that question on the manual instruction was taken in class. There were discussions about the use of buffer and what it meant to have low respectively high pH in a solution. When it was sorted out that high pH corresponded to high concentration of hydroxide ions the students could easier write the reaction formulae.

The students got experience of writing chemical formulae, used concepts like dissolve, solution, reaction and chemical names as cobalt sulfide and chromium hydroxide. This corresponds to the goal:

Pupils should be able to interpret, write and use the formula for chemical compounds and reactions, and thus reason in stoichiometric terms, as well as carry out simple calculations.

The students also got to see how redox reactions are written, which relates to the goal of the Chemistry A course:

[...] use the concept of oxidation and reduction, and describe applications in industrial and everyday contexts.

The student also got a recollection on the pH concept which is basic in the goal:

Pupils should have knowledge of the pH concept, neutralization, strong and weak acids and bases, as well as be able to discuss equilibrium in connection with e.g. cushioning effects and be able to relate this knowledge to environmental issues amongst others.

5.6 The students' behavior during the laboratory work

The students were in the beginning of Laboratory section II uncertain about how they should initiate the laboratory work. They were given the written manual instructions (where there was not an initial list of the material used, though all they would need was in the class room). They acted at the same time with curiosity, asking questions like:

Are we going to use the coned test tubes or the regular ones?

Why are the test tubes coned?

The students were uncertain about this because they found that both kinds of test tubes matched the centrifuge. This gave rise to a discussion of the advantage of having coned glass when you centrifugate; the precipitation is less scattered and stays on the side of the coned glass, this makes it easier to remove and separate it from the liquid. The students were also a bit puzzled about if they actually should use the laboratory clothing that had been put forward:

Shall we use the science students' coats?

referring to (ordinary) laboratory clothing¹ used by the students of the natural science program at the school. The students had earlier on only used plastic aprons during laboratory work. The teacher said that the students of course should use this laboratory clothing, especially since the laboratory work was more advanced than earlier work they had done and that it required this. This left them with a feeling that they were doing something out of the ordinary for students in their course. One student commented that:

It is fun doing something different (red. during the laboratory work).

The teacher experienced that they were more focused during the laboratory work, perhaps because they sensed that it was necessary and that it was something they had not done before (Section 5.8). After the laboratory work the teacher felt that the students seemed proud of what they had done. When some of the test tubes were shown during the recollection of the laboratory activities from last lesson, one student remarked that it (one of them) belonged to their group. These observations imply that the students understood that the teacher had high opinions of their abilities and their actions during laboratory work may have been influenced by the desire to show that they could cope with it.

¹Corresponding to the laboratory clothing used in the labs at the university basic courses in chemistry.

5.7 The teacher's role during the laboratory work

A teacher's role is not always to be in the center of attention. During the laboratory work the teacher walked around in the class room casually, curiously asking what step they were doing. I believe that she in this casual way managed to have full control of what the students were doing, without them feeling inspected at all times. If they hesitated in some steps she was not farther away than they could ask for her. She often told them that it was looking good, because it usually was. I believe the students felt confident in doing the laboratory work, knowing that the teacher was nearby.

The students for instance had not prior to the laboratory work used a centrifuge. When students were hesitant about where to put their test tubes, she did not jump in and tell them. She waited so that other students had an opportunity to think of an answer of their own.

In what slots should I put the test tubes?

Another student answered:

Opposite each other, of course!

The students agreed on this, realizing that the best thing would be that the centrifuge was evenly balanced. They also agreed on checking that they had not marked their tubes in the same way. In other case there could be a mix up.

The teacher awaited the student's questions instead of telling them on beforehand matters that she thought they would ask about. When questions

occurred the feeling was more towards "we will sort this out together" than "I know the answer". She reasoned with the students and at the same time conveyed calmness and assuredness. When one of the students opened the test tube she did not get angry she just remarked that it smelt funny in there. The students told her what had happened and she said there was a reason not to open the test tubes outside the fume-cupboards, which they now had discovered. She also said that it was a good thing that the contents smelt strongly, otherwise something might not have been done about it at once. The worst case being if there was an effluent of gas that had no smell, such as carbon monoxide. The teacher felt no need of getting annoyed. What was done was done and the students worked out the situation in short time, opening windows and corked up the test tube. Because of the smell no student would probably have tested to do this a second time anyway.

5.8 Interview with the teacher

The interview with the teacher was centered around the questions mentioned in Section 4.3, but came also to include the interview person's opinion on her role as a teacher during laboratory work.

What made the teacher interested in using these laboratory experiments in the teaching, is that she had a group of students who were interested in chemistry and that she believes it is important that the chemistry section in Science studies A gives something more than the chemistry taught in lower secondary school. The fact that the laboratory work was divided into separate laboratory sections was appealing. It is possible to start small

and then to do as much of the laboratory work as wanted, settle with the first laboratory section or go further. The teacher stressed that it is always fun to let the students carry out something more demanding than they earlier have encountered. This could initially make the students feel insecure, therefore it is important how you present the new task. The teacher described how she presented these laboratory work and asked them if they were interested. She made clear that this perhaps was a bit more difficult than earlier laboratory work the students had done, but that she believed that they could do it, and do it well.

The teacher saw the laboratory work as a suitable finishing of the chemistry section of the course. Earlier the students had identified anions by using spot plates. For instance had the students identified chlorides by adding silver(I) and gaining silver(I)chloride. They had also identified sulfate ions by mixing sodium sulfate and barium chloride. Since the students knew that sodium chloride is soluble in water they could establish the precipitation of barium and sulfate, that is barium(II)sulfate. The students had also identified carbonate ions by adding hydrochloric acid, when carbon dioxide was generated. To enhance the degree of difficulty and to provide something different, this laboratory work was represented to the students and, after their approval, carried out stepwise.

The carrying out of the laboratory work she believes has affected her teaching, in such a way that it provided suitable challenges for the students. Some students could feel elated at that they were able to accomplish this more difficult task, others could see the laboratory lesson as a possibility to show their abilities. The laboratory work included an *analysis*, normally

they had done synthesis. The students were presented with an analysis and also new techniques how to purify compounds. Earlier the students had only established that there was or was not a precipitation taking place on the spot plates. Now they got to centrifugate, wash and purify the precipitation and finally dissolve it. They were to interact with their created precipitations. In that way the challenge was greater than in the student's earlier laboratory work.

The teacher apprehended that the students' interest of the laboratory work was much higher than average for all students, not only from the normally most interested ones. She felt that the students put more effort into the experiments. During the laboratory work she noticed that students from different laboratory groups shared their observations which she much appreciated.

The teacher stressed that the courses in Science studies handle environmental issues, and that analyzing the presence of metal ions and nutritions could be a first step of environmental work. She mentioned an investigation that was done in the community, where they analyzed groundwater and soil.

The fact she wanted the students to remember from the laboratory work is that pH has a great impact on the solubility of metal compounds. This gives the students a better understanding of what can happen in nature when its exposed to acidification.

For the laboratory work to be more adaptable to the teaching in Science studies A, the changes she would have considered were to examine metal ions more common in the ground water and soil. Next time she may have put the laboratory work closer to the section concerning environmental

issues.

5.9 The practical work and instructions of the laboratory manual

The practical work and instructions of the laboratory work during the lessons (Section 5.3) are below analyzed according to the different aspects mentioned in Section 2.5.

The chemistry involved in the laboratory work included precipitation reactions and redox reactions combined with the use of pH and buffering systems. In the manual the students accounted for their observations during the experiments and wrote chemical formulae, the latter with some help from the teacher (Section 5.5).

There was no presence of any problems precipitating $PbCl_2$ after heating, which had occurred during the prelaboratory work, see Section 4.2. The arrangement with generating H_2S from FeS and HCl also worked out to satisfaction. There was no smell detected from the fume-cupboard when the gas was bubbled through the students' solutions. A suggestion is though to leave the equipment over night before doing anything about it. The equipment used in the testing was cleaned shortly after the laboratory lesson, and then it still contained some H_2S . Even though the bubbling (generation of gas) has stopped some time ago, the air in the equipment and in the surroundings contained enough gas to be detected by the smell.

The equipment was during the laboratory lesson put in a fume-cupboard in the preparation room next to the lab to separate it from the students'

classroom in case some gas had leaked out. The only time the gas was sensed by students was when one student on purpose uncorked a test tube outside his group's fume-cupboard. The smell was only noticeable a short time, thanks to the quick actions by the students (Section 5.3).

Some students used only ammoniac to make the chromium solution slightly basic, which worked out fine (Section 5.5). There were earlier on (in the prelaboratory work) concerns on the solution getting too basic, therefore the use of a buffer was suggested in the laboratory manual. Subsequently, according to Hägg [14, p. 653], a high concentration of ammoniac could give rise to ammin complex, such as $Cr(NH_3)_6^{3+}$, but in this case that did not occur. The students did not take too much ammoniac for this to happen. The use of only ammoniac could be added as an alternative in the manual to save time and decrease the amount of chemicals used. Some students got dichromate instead of chromate, probably due to not having made the solution enough basic (see Appendix B.6). This could be something to stress on during further implementations of the laboratory work.

The time allocated for Laboratory section I and III studied in the testing was enough for implementing the laboratory work. The first section only took about half an hour, the same for the testing of unknown samples. In the third section the students just had time to finish the laboratory work. Perhaps some more time would have been preferred because of the cleaning up was done in a bit of a haste. Perhaps it also would have been good to have some time talking about the laboratory work with the students shortly afterwards. The 20 minutes lost in the beginning of the laboratory lesson would probably have been sufficient for that purpose (see Section 5.3). The

recollection was done the next lesson instead. This gave on the other hand the students a possibility to think about the implemented laboratory work on their own. It would have been nice to have had a whole afternoon or day to do all laboratory work and to also have tested it in classes of Chemistry A (see Section 2.6), because much of the theory in the laboratory work relates to this course and was in the beginning the prime target group.

Chapter 6

Discussion and Conclusions

This chapter is divided into different three sections corresponding to the parts of the problem statement in Chapter 3. The *learning potential* of the laboratory work is discussed in terms of practicability, concept learning and as a motivating factor in science learning. The students' *self-efficacy* before, during and after the laboratory work is discussed as well as the need for (and the different types of) *collaboration* occurring in the laboratory work.

6.1 Learning potential

A basic condition that can be required for laboratory work to be meaningful is that the implementation is practicable; further, the laboratory work should produce understandable results. The students gained from the discrepant results because the latter discussion gave new insights into chemical equilibrium. In the present laboratory work the students did not get a direct answer why some students got a yellow precipitation and others got an orange precipitation in one of the parts of the laboratory work. This

created a kind of situation that Berg [8] considers as being inspiring for young students. Berg believes that, in this kind of situation, the students would start discussing questions to which there is no clear given answer. In this thesis an implementation of the laboratory work was made in the course Science studies A, which typically has an environmental focus (Section 2.6). The teacher participating in this study considered the laboratory work, presented in this thesis, as an excellent opportunity to connect practical elements with the theoretical aspects on pollution. The teacher also regarded this laboratory work as a good opportunity to rehearse the pH-concept. During the laboratory work the students saw how pH influences its surroundings. This can be compared with most of the students prior to this laboratory work only had measured pH in some solutions and confirmed its existence. In connection with the laboratory work the students also trained on writing chemical formulae and used concepts such as reaction, solution, and precipitation. The students learned the basics of chemistry, such as having a lucid picture of these latter concepts, to get more interested in environmental issues. The main goal of the laboratory work in the course Science studies A was to show the dynamic properties of nature. Changing the pH of a solution could have a large effect on the reactions within. What is done to nature by man has after-effects. If you in advance learn more about chemical reactions and what drives them, you can use it in a constructive way. This is what is actually done in class, using chemical theory for analytical gain. The chemistry involved can be measured in different degrees of difficulty. When it comes to chemical theory the over-all picture (presented above) is the most important knowledge. The theory re-

garding chemical equilibriums, driving forces for reactions, and calculations on redox reactions (Section 2.4) is included in the goals of Chemistry A (Section 2.6).

Being aware of the origin of the laboratory work students realized that they were confronted with a challenge. The teacher experienced that the students tightened up—due to the requirements of the situation—during the laboratory work. That is, the students worked more methodically and with higher accuracy compared to how they usually work during laboratory sessions. The students made observations about physical changes such as color variations, state of aggregation etcetera. Based on their observations they also made analyses. This gave the students a clear purpose of the experiments. As Pickering [21] states as important the students not only observed but drew conclusions and used this to go on with the analysis. Sjøberg [23] states that a scientific observation should be accompanied by the use of language and theory consistent with science. The written notes from the students' analyses show that many of the students used chemical concepts to describe their laboratory work, therefore training to see chemical reactions from a more scientific view. Taking Pickering's thought one step further would be to think about the limitations of the laboratory work. The students were told about the precipitation reagent that would make it possible to separate the contents of different samples. One adjustment could be not to tell the students that for instance heating would separate the different ions in this certain case. This is a way of making the laboratory work more of the students own doing. Letting the students in conclusion make analysis worked as a motivating factor for the implemen-

tation of the laboratory work. The challenge for the teacher is to adjust how much information that should be given from start.

6.2 Collaborations

The laboratory work gave rise to collaboration between students, laboratory groups and the teacher. Most students felt that they needed collaboration to be able to manage the tasks, which is in line with what Vygotsky would call learning through collaboration. This was expressed in different ways: by asking other groups of their results, discussions on how to use different equipments among students, and discussions with the teacher on chemical theory. Different types of collaboration also occurred within the laboratory groups. Some groups divided the work equally, doing every step together, others gave each other different roles, as in the studies by Eskilsson and Helldén[11]. It would have been interesting to look further trying to correlate types of collaboration with concept learning. In one of the steps in the implementation of the laboratory work, the lab groups changed. Students that had not participated in the initial experiments took on other students. In this case the more experienced students did not mind taking charge of the group work. The other students listened carefully and managed to make up lost experience, at least for the ones necessary for the experiments. The 'new' students thereby welcomed the collaboration with the 'older' students. In the study by Eskilsson and Helldén[11] the students (though younger than the student in this study) were of the opinion that explaining to each other and making discussions are the main reasons for their understanding. The teachers in Eskilsson and Helldén's study were

very positive towards laboratory work and like in this study worked as interested conversation partners. They did not give the answers but wanted to start discussions between the students and between the students and the teacher. Similarities can be seen in the observation done in this study (Section 5.3).

6.3 Self-efficacy

It serves a purpose to clearly indicate the origins of the laboratory work. The teacher presented the laboratory work as an opportunity for the students to do something out of the ordinary. The fact that all students managed to carry out the laboratory work gave them a feeling that they had succeeded in doing something more demanding than their usual tasks. Prior to the laboratory work many students did not have high opinions of their knowledge in chemistry, the recollection after the chemistry section and the laboratory work showed otherwise. To reach such a result could imply that the self-efficacy has enhanced, but also that the students felt that they had advanced their learning in chemistry. Bandura [7] stresses that accomplishing something that is considered hard develops an interest in the subject. Further, educational practices, in this case a kind of laboratory work, should not only be graded by the skills and knowledge, but also what they do to children's beliefs about their capabilities. After the laboratory work the students felt proud of their achievements. Both I and the teacher got the feeling that the students considered the laboratory work an interesting challenge and that everybody learned something new.

The resulting laboratory manuals and the work carried out by testing

class show that it is possible to redesign university level laboratory work to suit the upper secondary school teaching. Moreover, the students consider the laboratory work as a challenge and increase their performance level accordingly. Hence it is important to give the students challenging, but practicable, tasks to boost their self-efficacy.

Chapter 7

Consequences for the teacher's role

The students' and teacher's manuals are to be available for teachers and others interested in laboratory work in school. This thesis gives an insight into different aspects to consider when designing laboratory work for upper secondary school and how they can be treated. The manuals could contribute to the teaching and also give ideas of how teachers could create and design their own laboratory work. In this thesis the collaboration during laboratory work became an interesting factor. The different ways that were observed about how students collaborate and divide laboratory work can be used as a platform to easier analyze and reflect on the events during laboratory work. This could start discussions on how to put together lab groups in order to get everybody participating and learning the most from the situation. This thesis also shows an example where the teacher puts her faith in her students and gives them a challenging task. A task the students perhaps would not have thought they could manage to do. The

observations entail that teaching is about being courageous, and testing boundaries. The will to make changes and try new possibilities must be a driving force, since teaching should be and always will be a craft under development.

Appendix A

Students' manual



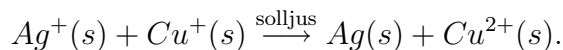
A.1 Fakta om metallerna och deras tillämpningar

Silver

Silver angrips inte av syre vid atmosfärstryck—detta är en egenskap hos ädelmetallerna dit även silver räknas. Det krävs mer än 15 bars tryck för att bilda oxiden Ag_2O . Silver används i bland annat mynt, bordsilver och silversmide på grund av sin beständighet, dock är metallen mjuk och legeras därför alltid (vanligen med koppar) då den används.

Ett stort användningssområde för silver är som komponent i fotokemiska material¹. Fotokromiskt glas är ett exempel på ett fotokemiskt material och används exempelvis i fönster, växthus och solglasögon. När solen ligger på mörknar glaset och förebygger att så mycket ljus tränger genom.

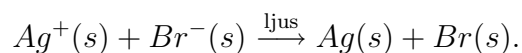
För att tillverka fotokromiskt glas används bland annat silvernitratt och kopparnitrat som smälts samman med vanligt glas. För att få silver- och kopparjonerna att kristallisera tillsätts även en metallhalid. Vid starkt ljus reduceras silverjonerna till silver, vilket får glaset att mörkna. Samtidigt oxideras koppar(I)jonerna till koppar(II),



Eftersom koppar(II)jonerna inte har någonstans att ta vägen kan reaktionen återgå när ljusintensiteten avtar. Denna praktiska tillämpning av fotokemi har sina rötter inom fotografering. För att skapa svart-vita fotografier belyses en plastfilm täckt av mikroskopiskt små silverbromid-

¹Fotokemiska material är material i vilka det sker reaktioner då de utsätts för ljus.

kristaller inbäddade i gelatin.



Efter att filmen behandlats så att avtrycket blir bestående erhålls ett film-negativ. För att framkalla fotografiet sätts negativet mellan en ljuskälla och en lins. Avbilden fångas upp på ett ljuskänsligt papper täckt med silverklorid. Efter att pappret ytbehandlats för att inte vidare reagera med omgivningen är det ett färdigt fotografi.

Lösliga silverföreningar är giftiga, men har specifika medicinska tillämpningar. Exempelvis används silverjoner som bakteriedödande ämne inom sjukvården. Spädbarns ögon baddas med en mycket utspädd lösning av silvernitrat, $AgNO_3$, för att förebygga infektioner.

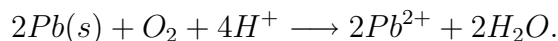
Bly

I naturen förekommer bly oftast som mineralen galena. För att utvinna bly ur mineralen rostar den och omvandlas då till blyoxid, PbO . Genom att upphetta oxiden tillsammans med kol skapas bly. Blyföreningar brukar även finnas bland silvermalm och ses som en biprodukt vid brytningen av den mer ädla metallen.

Bly hör till tungmetallerna, och är ansedd att vara den giftigaste metallen. På grund av sin höga densitet ger bly ett bra skydd mot radioaktiv strålning. Metallen är mjuk men ger ett bra skydd mot korrosion. I luft bildas ett skyddande oxidskikt på metallen.

Många blyföreningar har en sötaktigt doft och smak. Denna egenskap tog romarna till vara då de använde bly för att smaksätta alltför bittra drycker. Många kärl brukades även ge ett skikt av bly för att just motverka korrosion. I början skyddas blyet av passivering, antingen av oxid, sulfat eller fostatskikt, men vid god syretillgång löses skiktet upp och blyjoner frigörs.

Äldre typ av metallburkar (33cl) hölls samman av lödtenn, som är en blandning av tenn och bly. Om man exempelvis lät frukt juice eller annan sur dryck stå öppen, så frigjordes blyjoner i drycken/vätskan. När burken öppnas händer följande reaktion:



Detta sker även i gamla vattenledningar om vattnet är tillräckligt surt och mjukt. Då är det klokt att låta kranen spola ett tag om vattnet stått stilla under natten. Om vattnet istället är hårt, dvs innehåller en hög halt karbonatjoner, uppstår inte problemet. Då bildas ett skyddande skikt av

A.1. FAKTA OM METALLERNA OCH DERAS TILLÄMPNINGAR 91

blykarbonat ovanpå blyet. Blyförgiftning var tidigare ett stort problem i Storbrittanien. Britternas lösning är att tillsätta fosfatjoner i vattnet och därmed få ett skyddande skikt av blyfosfat.

Bly är en mjuk metall med en förhållandevis låg smältpunkt för att vara en metall, 327.4°C . På grund av detta användes bly under Antiken för att forma rörledningar. Blyförgiftning medför vid låga koncentrationer huvudvärk men kan vid grav förgiftning ge njursvikt och hjärnskador.

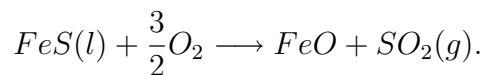
Bly används även i ammunition. Skadeskjutna djur kan bli så sjuka av blyförgiftning att de dör. Vissa djur äter upp hylsorna och på så vis kommer metallen in i kretsloppet. Även fåglar som äter kadaver får i sig bly. På grund av detta har blyammunition förbjudits i många länder.

Många blysalter ger stabila starka färger. Förr användes användes dessa som färgämne i godispapper, tidningar och matprodukter. En förening kallad "vit bly" användes som inomhusfärg. Det finns uppgifter om att barn börjat äta på färgflagor från väggarna på grund av dess söta smak. Idag används istället exempelvis titanoxid för att få vita väggar, vilket inte är lika giftigt eller sött.

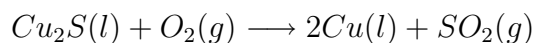
Koppar

Av världens kända koppartillgångar utgörs 90 procent av sulfidiska malmer, och endast 1 procent av gedigen koppar. De återstående procent är i form av oxider. Många hustak som är belagda med kopparplåt får med tiden en grönaktig beläggning kallad ärg. Detta beror på att kopparn reagerar i fuktig luft eller vatten och oxideras till kopparsalter.

I Norden är kopparkis, $CuFeS_2$, den mest förekommande mineralen. Det finns ett antal olika tekniker för att utvinna kopparn. För att ta tillvara kopparn kan man exempelvis smälta ner mineralen och då anrika den i ett sulfidskikt av CuS och FeS . För att skilja koppar och järn åt utnyttjas att koppar har en högre affinitet för svavel än vad järn har. Detta betyder att koppar binder hårdare till svavel än vad järn gör. Genom att blåsa syrgas genom sulfidskiktet bildas järnoxid, som kan separeras bort,



Kvar blir en sulfidfas (ett skikt med sulfider) med betydligt högre kopparhalt. Genom ytterligare blåsning av syrgas reagerar även kopparsulfiden,



och man får flytande koppar. Koppar används mycket i legeringar, såsom mässing, nysilver och brons och har stor användning i elektriska ledningar på grund av sin höga konduktivitet. Ren koppar har den högsta termiska konduktiviteten bland alla metaller. Därför används den ofta i köksutrustning för att snabbt värma upp botten av kastruller. Koppinoxid, Cu_2O , används i bottenfärger för båtar för att förhindra beväxning av djur och

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växter. Koppar används även i fotokromiskt glas beskrivet i avsnittet om silver.

Förr användes kopparkärl vid matlagning. Det hände då att koppar löstes från kärlen och man kunde ibland få kopparförgiftning. Numera inträffar kopparförgiftning oftare till följd av att koppar löses ut från kopparrör som används i vattenledningar. För att slippa dessa besvär kan man spola några liter vatten innan man tar vatten. Då sköljs eventuellt upplöst koppar bort. För människan är koppar och dess föreningar måttligt giftiga. Svårare förgiftningar är sällsynta eftersom man på ett tidigt stadium blir illamående och får kräkningar. Besvär uppstår vid halter av 1-3 mg koppar per liter vatten.

Samtidigt ingår koppar naturligt i biologiska system hos både människor och djur. Den ingår i vissa proteiner liknande enzym och i levern finns en kopparreserv i form av en kopparhaltig blå globulin kallad ceruloplasmin. En brist på koppar leder till att kroppen inte kan tillgodose sig det järn som är lagrat i levern. Hos ryggradslösa djur finns kopparproteiner kallade hemicyanin, som är syrebärare precis som hemoglobin. Koppar är den tredje viktigaste övergångsmetallen näst efter järn och zink. Som människa behöver man få i sig 5 mg koppar per dygn.

Kobolt

Kobolt finns som mineral i vår jordskorpa. Där kobolt oftast i form av sulfider eller arsenider. Framställningen av kobolt är en svår process eftersom att den malm som bryts oftast innehåller ett flertal metaller, såsom järn och nickel, ibland även koppar och silver. För att utvinna kobolt används en process där malmen smälts ner, skiktas och senare behandlas med svavelsyra.

Kobolt-60 ger vid β -sönderfall upphov till en intensiv gamma-strålning med hög energi och har en halveringstid på 5,2 år. Isotopen framställs från naturliga kobolt-59 genom neutronbestålning i en kärnreaktor och används inom det medicinska området för strålbehandling av vissa typer av cancer.

Kobolt har en mycket viktig biologisk funktion som beståndsdel i vitamin B_{12} . Vitamin B_{12} är en del av vitamin B komplexet och hittas huvudsakligen i kött och mejeriprodukter. Vitaminen är nödvändig för framställningen av röda blodkroppar, återuppbyggnaden av nervsystemet samt barns tillväxt och utveckling. En brist på B_{12} kan leda till ameni (blodbrist). Detta yttrar sig genom extrem trötthet, andnöd och dålig motståndskraft för infektioner. Kobolt spelar en enzymatisk roll i kroppen. Figur A.1 visar hur koboltföreningen är inblandad i framställningen av DNA.

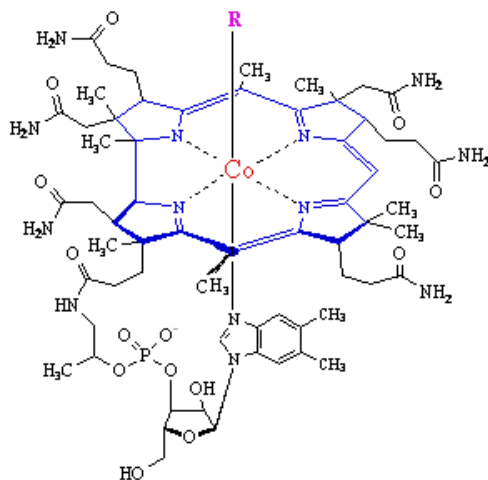


Figure A.1: Längst ner i figuren sitter en kvävebas bunden till metalljonen samtidigt som en del av dess porfyrinlikande ring håller fast i en fosfatgrupp. Fosfatgruppen är bunden till deoxiribos precis som i DNA.

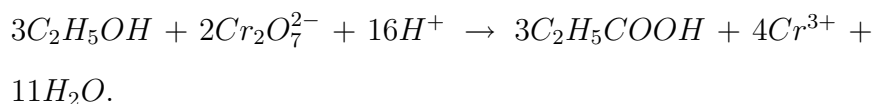
Krom

Krom återfinns i vår jordskorpa och då ofta som kromit, $FeCr_2O_4$. Vid upphettning i luft bildas på metallen en skyddande hinna av kromoxid, CrO_3 , vilket förebygger senare angrepp på metallen. Denna egenskap hos krom är anledningen till att metallen används som legeringselement i stål för att ge beständighet mot korrosion, därmed blir stålet 'rostfritt'. En annan fördel är att krom gör stålet hårdare. Namnet krom kommer av ordet "chroma" som betyder färg. Många kromföreningar har väldigt distinkta färger. Detta tas tillvara inom färgindustrin men har även fått en del udda tillämpningar. Exempelvis används dikromat för att mäta promillehalten alkohol vid inandningsprov. I den norska versionen av Kemiolympiaden²

²En årlig tävling för gymnasieelever, som även finns i Sverige.

2005 fick eleverna begrunda en redoxreaktion hämtad från verkligheten.

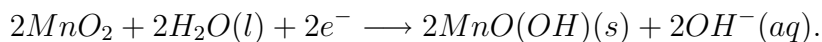
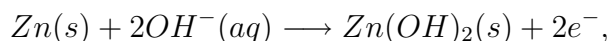
Når politiet mistenker en bilfører for promillekjøring, må vedkommende ta en såkalt alko-test, som innebærer at alkohol i pusten vil føre til en fargereaksjon i alkometeret. Metoden er basert på at etanol oksideres til eddiksyre når luften ledes gjennom en sur løsning av dikromat, og $Cr_2O_7^{2-}(aq)$ [orange farge] reduseres til $Cr^{3+}(aq)$ [grønn farge]. Likningen for denne reaksjonen er



Dikromat som används i exemplet ovan har oxidationstal VI och är därmed en förening som anses carcinogen. Samtidigt som det inte vore bra att komma i kontakt med krom(VI)föreningar behöver vi små mängder av krom(III) i vår kost. Insulin tillsammans med krom(III) har en högst betydande roll i att reglera glukosnivåerna i blodet. En brist på krom(III) kan långsiktigt leda till diabetes.

Mangan

Mangan finns ganska rikligt i jordskorpan. Viktigast som malmer är oxider och oxidhydroxider, bland dem särskilt brunsten, MnO_2 , och manganit, $MnO(OH)$. De gamla grekerna använde manganföreningar för att avfärga glas, vilket även gav dem namn efter deras ord för att rena, manganes. Innan nickel-kadmium batterierna lanserades, var alkaliska batterier baserade på zink och brunsten mycket vanliga. Ett zinkhölje utgjorde anoden och en stav av grafit och brunsten var katod. Som elektrolyt, lösning som kunde leda ström, användes kaliumhydroxid. I cellreaktionen oxideras zink till zinkhydroxid och brunsten reduceras till manganit,



Vid nettoreaktionen produceras lika mycket hydroxidjoner som det konsumeras, vilket var en stor fördel jämfört med tidigare typer av batterier.

Mangan har en betydligt större tillämpning inom biokemi. Tidigt insåg man att små mängder mangan var nödvändigt för gröna växters klorofyllbildning, även om det till synes inte ingår i klorofyllmolekylen. Mangan upptas av växterna i form av $Mn(II)$. Ju mer basisk miljö desto mer benägen är $Mn(II)$ att oxideras till $Mn(IV)$, vilket kan ställa till problem för växterna. På basiska jordar visar därför växterna tecken på manganbrist, ofta i form av gråa eller ljusa fläckar på bladen, vid betydligt högre manganhalter än på surare jordar. Detta brukar avklaras med att surgöra jordarna. Mangan ingår även i djurs enzymsystem, bland annat återfinns det i leverenzymer arginas, som gör om kvävehaltiga restprodukter till urinämne.

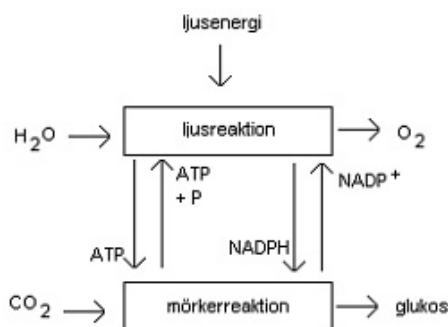
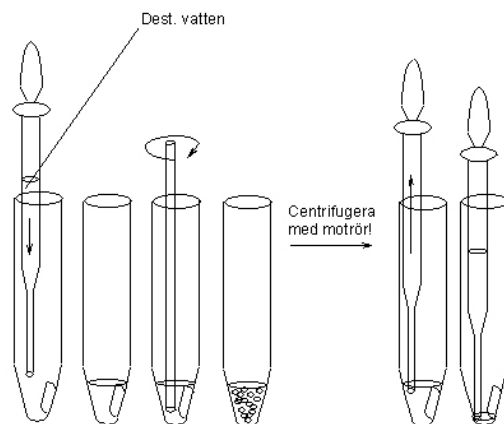


Figure A.2: Schematisk bild över fotosyntesen.

Inom senare forskning har manganets koppling till klorofyll och växters fotosyntes tydligare kartlagts. Som många övergångsmetaller är manganets roll att agera cykliskt som både reduktions- och oxidationsmedel. Genom att sönderdela vatten till syrgas frigör ett mangankomplex elektroner, som sedan används för att bygga upp den energiladdade molekylen NADPH. Denna tillsammans med ATP som också bildas i ljusreaktionen, används för att driva mörkerreaktionen. Produkten i mörkerreaktionen är glukos-cellernas bränsle.



A.2 Laborationsteknik: Tvättning av en fällning

När man ska separera en fällning från en lösning är det svårt att suga upp all vätska. Vätskan innehåller ofta diverse joner som man inte vill skall förorena det fasta ämnet. För att försöka få en fällning som endast omgärdas av rent vatten, späder man därför ut den vätska som då blir kvar. Om exempelvis 0.1 ml vätska är kvar och denna späds till 1 ml, finns det färre metalljoner per volym vätska. I den 0.1 ml vätska som blir kvar efter nästa uppsugning, finns då förhoppningsvis en tiondel så många metalljoner! Om man gör detta flera gånger blir sannolikheten att det endast finns kvar några få metalljoner i vätskan väldigt stor.

Tvätta en fällning innebär

- att man centrifugerar så att fällningen samlas nederst i röret
- att man suger bort vätskan
- att man fyller på med dest.vatten, lika mycket som man tagit bort
- att man rör om i provröret med glasstav så att fällningen tvättas

- att man centrifugerar igen
- att man slutligen försiktigt tar bort vätskan ovanför fällningen

Nu är fällningen tvättad!

A.3 Jonlaboration I

Om du blandar två lösningar och det bildas en fast förening, så kallas denna förening en fällning. Det du gjort är en utfällningsreaktion. Laborationen går ut på att skilja två metalljoner åt genom att fälla ut den ena jonen medan den andra är kvar i lösningen.

Silver och Bly

Silver bildar i vattenlösning envärda joner, Ag^+ , och bly tvåvärda, Pb^{2+} . Silver(I) och bly(II) bildar båda kloridföreningar när de blandas med saltsyra. Det går att skilja dem åt om man värmer kloridfällningarna. Då löser sig endast en av dem.

Förberedelser

Fyll en 250 ml bägare till hälften med vatten och ställ på en kokplatta (eller på trefot och gasolbrännare). Låt vattnet koka upp.

Till tvättning av fällningarna behövs en glaspipett och en dest.vattenflaska.

Metod

Droppa 1 ml $AgNO_3$ i ett rör.

Till samma rör tillsätt 3 ml 0.5M saltsyra.

Vad händer?

Droppa i ett annat rör 1 ml $Pb(NO_3)_2$.

Tillsätt 3 ml 0.5M saltsyra.

Vad händer?

Centrifugera båda rören.

Vilken färg har det som bildats?

Vad har bildats?

Tvätta fällningarna 3 gånger enligt beskrivningen ovan.

Efter tvättningarna fyll på med 1 ml dest.vatten i varje rör.

Ställ rören i vattenbadet.

Rör om i varje rör med var sin glasstav.

Vad händer?

A.4 Jonlaboration II

Om du blandar två lösningar och det bildas en fast förening, så kallas denna förening en fällning. Det du gjort är en utfällningsreaktion. Laborationen går ut på att skilja två metalljoner åt genom att fälla ut den ena jonen medan den andra är kvar i lösningen.

Koppar och bly

Både Cu^{2+} och Pb^{2+} bildar sulfider i sur lösning, men det krävs olika starka syror för att lösa upp respektive sulfid. Sulfidjoner, S^{2-} , till reaktionerna fås genom att lösa upp järnsulfid (FeS) i saltsyra. Då bildas en ofärgad gas, svavelväte, H_2S . Detta utförs av läraren vid angiven plats.

Förberedelser

En buffert är en vätska som ser till att surheten på lösningen inte ändras då en reaktion sker. Tillverka en pH2-buffert genom att blanda 1 ml Na_2SO_4 och 1 ml $NaHSO_4$ i ett provrör.

Metoddel 1

Ni kommer att undersöka en lösning med kopparjoner som ni bearbetar under de instruktioner som anges med ett A parallellt med en lösning som innehåller blyjoner som studeras under B.

A. Häll 1 ml kopparnitrat $Cu(NO_3)_2$ i ett provrör.

Sätt till 1 ml pH 2-buffert.

Observationer: _____

Vad har bildats?_____

B. Häll 1 ml blynitrat $Pb(NO_3)_2$ i ett provrör.

Sätt till 1 ml pH 2-buffert.

Observationer: _____

Vad har bildats? _____

A och B. Gå till dragskåpet där gasflaskan med svavelväte finns och låt läraren bubbla ner gasen i era provrör.

A. Observationer: _____

Vad har bildats? _____

B. Observationer: _____

Vad har bildats? _____

Reaktionsformel i A: _____

Reaktionsformel i B: _____

Metoddel 2.

A och B. Tvätta fällningarna med 1 ml dest.vatten och 2 droppar pH2-buffert till varje provrör. Centrifugera båda rören.

A. Fortsättning

Sug bort vätskan och häll 1 ml dest.vatten i provröret med fällningen. Till-sätt 5 droppar 8M salpetersyra.

Observationer: _____

Om inget händer, sätt provröret i ett vattenbad. Rör om under uppvärmn-

ing.

Observationer:_____

B. Fortsättning

Sug bort vätskan och häll 1 ml dest.vatten i provröret med fällningen. Tillsätt 5 droppar 8M salpetersyra.

Observationer:_____

A och B. Om inget händer, sätt provrören i ett vattenbad. Rör om under uppvärmning.

Observationer:_____

Avslutning

I vilket av fall A och B kunde fällningen lösas upp med hjälp av en stark syra?

Ange reaktionsformel för den upplösta fällningen.

Extrauppgift

I provröret där du fick upplösning, tillsätt några droppar koncentrerad svavelsyra. Centrifugera!

Observationer:_____

Vad har bildats? _____

Reaktionsformel: _____

A.5 Jonlaboration III

Om du blandar två lösningar och det bildas en fast förening, så kallas denna förening en fällning. Det du gjort är en utfällningsreaktion. Laborationen går ut på att skilja två metalljoner åt genom att fälla ut den ena jonen medan den andra är kvar i lösningen.

Krom och kobolt

När trevärt krom, Cr^{3+} , och tvåvärt kobolt, Co^{2+} , låts reagera med sulfidjoner och hydroxidjoner i basisk lösning bildar den ena en sulfidförening och den andra en hydroxidförening. Genom att göra lösningarna lite surare löses hydroxidföreningen upp igen, men däremot inte sulfiden. Sulfidjoner, S^{2-} , till reaktionerna fås genom att lösa upp järnsulfid (FeS) i saltsyra. Då bildas en ofärgad gas, svavelväte, H_2S . Detta utförs av läraren vid angiven plats. Väteperoxid används under laborationen som oxidationsmedel. I detta fall får väteperoxiden en metalljon att öka sin laddning åt det positiva hållet. I detta fall går trevärt krom, $Cr(III)$, till sexvärt, $Cr(VI)$. Själv så sönderdelas väteperoxiden till vatten och syrgas.

Förberedelser

En buffert är en vätska som ser till att surheten på lösningen inte ändras då en reaktion sker. Tillverka en pH2-buffert genom att blanda 1 ml Na_2SO_4 och 1 ml $NaHSO_4$ i ett provrör. Tillverka en pH9-buffert genom att blanda 1 ml NH_3 och 1 ml NH_4Cl i ett provrör.

Metoddel 1.

A. Häll 1 ml kromnitrat $Cr(NO_3)_3$ i ett provrör.

Sätt till 1 ml pH 9-buffert.

Observationer: _____

Vad har bildats? _____

B. Häll 1 ml koboltnitrat $Co(NO_3)_2$ i ett provrör.

Sätt till 1 ml pH 9-buffert.

Observationer:_____

Vad har bildats? _____

Gå till dragskåpet där gasflaskan med svavelväte finns och låt läraren bubbla gasen i det provrör där inget har hänt.

Observationer:_____

Vad har bildats?_____

Vilken av jonerna bildar sulfid i basisk lösning och vilken bildar hydroxid?

Skriv reaktionsformler för respektive fall.

Reaktionsformel i A:_____

Reaktionsformel i B:_____

Metoddel 2.

A och B. Tvätta fällningarna med 1 ml dest.vatten och 2 droppar pH9-buffert till varje provrör.

Centrifugera båda provrören.

A. Fortsättning

Sug bort vätskan och häll 1 ml pH2-buffert i provröret med fällningen.

Observationer:_____

Tillsätt några droppar 5M NaOH till så att lösningen blir starkt basisk (pH 14). Kontrollera med pH-papper.

Tillsätt sedan försiktigt 2 droppar väteperoxid, H_2O_2 .

Teckna reaktionen då krom(III) och hydroxidjoner reagerar med väteperoxid för att bilda kromatjoner, CrO_4^{2-} .

Reaktion:_____

Värm lösningen på vattenbad så att överskottet av väteperoxiden sönderdelas. Avsluta värmningen då gasutvecklingen avtagit. Ta upp provröret ur vattenbadet. Tillsätt någon droppe $Pb(NO_3)_2$.

Observationer:_____

B. Fortsättning

Sug bort vätskan och häll 1 ml pH2-buffert i provröret med fällningen.

Observationer:_____

Centrifugera.

Sug bort lösningen i provröret.

Tillsätt 7 droppar dest.vatten och 7 droppar konc HNO_3 för att lösa upp fällningen.

Observationer:_____

A.6 Referenser

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Appendix B

Teachers' manual



B.1 Inledning

Hej!

I dina händer håller du nu en lärarhandledning som utgör en del av mitt examensarbete inom kemi med didaktisk inriktning. Laborationen har genomförts i en gymnasieklass inom deras undervisning i Naturkunskap A. Detta var för dem en positiv upplevelse, då de fick göra något utöver det vanliga ¹. Jag ser laborationen som en möjlighet att koppla ett laborativt moment till kursens studier om metallföreningar och deras koppling till miljön. Den bakomliggande kemin behandlar pH-begreppet och dess inverkan vid kemiska reaktioner. Fokus ligger på ett antal metaller och deras föreningar. Laborationen är tänkt som ett större projekt med dellaborationer, men varje del kan mycket väl användas som fristående element i undervisningen.

Trevlig läsning!

¹Mer att läsa finns i mitt examensarbete: Laboratory Chemistry in Natural Science

B.2 Syfte och mål relaterat till

Naturkunskap A

Bland kursmålen till Naturkunskap A står det skrivet att

Eleven skall kunna göra observationer och enkla experiment
samt kunna analysera och tolka resultaten.

Samtidigt som ämnet Naturkunskap skall bygga på elevernas tidigare erfarenheter och kunskaper från grundskolan, är det viktigt att gå ett steg till – att lyfta upp undervisningen till en annan nivå. Enklare jon-laborationer har eleverna oftast redan stött på i grundskolan ². I denna nya tappning tar de till vara denna kunskap men får se den i ett nytt perspektiv. De ser sin kunskap användas som en del av en naturvetenskaplig tillämpning. Må hända är det inte varje dag man just vill separera exempelvis krom och silver, men det visar på hur kemi kan användas på ett kreativt sätt. Laborationen kan därmed hjälpa till att *skapa förståelse av naturvetenskapens arbetssätt*, vilket ämnet även syftar till per definition enligt Skolverket.

Ämnet Naturvetenskap är ett tvärvetenskapligt ämne där naturvetenskapliga frågeställningar kan studeras ur flera perspektiv

Naturkunskap A har en utpräglad inriktning mot miljöfrågor. När det gäller kemi finns det mycket att orda om angående olika föreningars giftighet och deras inverkan på miljön. Om vi lär oss lite mer om kemisk reaktion-slära, kan man lättare utveckla angreppssätt för att lösa de miljöproblem som finns.

²Laborationer där eleverna exempelvis fyller i ett protokoll mellan vilka jonslag det uppstod fällning

Karakteristiskt för ämnet är blandningen av teoretiska studier, observationer, experiment och fältstudier.

Vad händer egentligen i en sjö som är utsatt för förorening? Vad händer med metallföreningarna i vattnet då pH ändrats? Vilka former är de mest skadliga? Detta är frågor som kan diskuteras i anknytning till laborationen och kan sammanlänkas med faktiska fältstudier. En strävan är att eleverna skall inse att pH har en viss betydelse för miljöns utseende.

Ett annat mål inom Naturkunskap A är att

Eleverna skall kunna beskriva miljöproblem utifrån studieinriktning.

Den förberedande teorin till laborationen siktar mot att behandla det kemiska innehållet utifrån det perspektiv eleverna kan känna igen sig i och har ett intresse kring. Naturkunskap A är en kurs som alla elevklasser läser oavsett inriktning. Därmed är det en balansgång att försöka skraddarsy en kemisk bakgrund som känns relevant ur alla perspektiv. I elevhandledningen ges olika exempel på vad som kan tas upp. Varför inte ge eleverna som arbetsuppgift att finna information? Det finns nog en stor sannolikhet att de fakta eleverna fastnar för har relevans för deras studieinriktning. Jag ser detta som ett lärorikt tillfälle både för dig och dina elever. Elevernas tillskott kan du dra nytta av i din strävan att ytterligare programanpassa din undervisning.

B.3 Syfte och mål relaterat till Kemi A

Det finns många av kursmålen som kan relateras till denna laboration, vilket är en del av dess styrka. Laborationen förenar många av de områden som behandlas i Kemi A. Bland bakgrundsfaktan som tillhör laborationen visas exempel på förekomst och tillämpningar i samhället. På detta sätt får eleverna en verklighetsanknytning till den kemi de behandlar i klassrummet.

Eleven skall ha kännedom om några grundämnen, kemiska föreningar och moderna material, deras egenskaper förekomst och kretslopp samt deras betydelse t ex i jordskorpan eller inom olika verksamhetsområden i samhället.

Kopplat till laborationen arbetar eleverna med att definiera och balansera de kemiska reaktioner som äger rum. Därmed får de erfarenhet av att

tolka, skriva och använda sig av formler för kemiska föreningar och reaktioner och därvid föra stökiometriska resonemang samt utföra enkla beräkningar.

Laborationsdelen innefattande kromföreningar ger ett exempel på en reaktion som innefattar oxidation och reduktion, samtidigt som det ges ett förslag på tillämpning i samhället. Detta satisfierar med målet att eleverna skall kunna

använda begreppen oxidation och reduktion och beskriva tillämpningar i industriella och vardagliga sammanhang.

Grundstommen i laborationen är kemiska reaktioner som berör löslighetsjämvikter och deras koppling till pH. pH-begreppet blir därmed centralt och eleverna får se buffertverkan i praktisk tillämpning. Eleverna behandlar även syror under laborationen. Mycket av detta ingår i det sista målet i Kemi A där eleverna skall

ha kunskap om pH-begreppet, neutralisation, starka och svaga syror och baser samt kunna diskutera jämvikter i samband med t ex buffertverkan och kunna relatera dessa kunskaper till bland annat miljöfrågor.

Laborationen behandlar en stor del av den kemi som förekommer inom Kemi A. Därför skulle den passa ypperligt som en repetition av de olika avsnitt som behandlats under kursen. Laborationstillfället kan ses som en möjlighet för eleverna att använda och befästa den kunskap de har tillförskaffat sig eller att se begreppen ur ett mer tvärvetenskapligt perspektiv.

B.4 Beskrivning

Laborationen utgår från ett separationsschema, illustrerat i Fig B.1, där metallföreningar skiljs åt med hjälp av pH-ändringar och varierad löslighet. Eleverna får genomföra ett antal karakteristiska reaktioner för att träna upp tekniker att ur ett prov separera olika metalljoner. Därefter kan eleverna få ett okänt prov som de skall bearbeta efter det mönster som de tidigare har studerat. Metalljonerna är indelade i tre grenar, där varje gren kan ses som en fristående laboration. Varje laboration kan sedan följas upp av att eleverna får ett prov med en eller flera av grenens metalljoner.

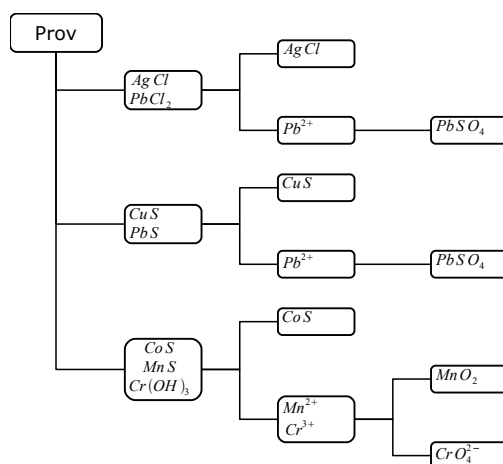


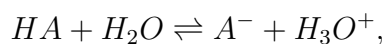
Figure B.1: Flödesschema över laborationen

B.5 Teori

Reaktionerna i laborationen behandlar utfällningsreaktioner, men även protolysjämvikter, komplexbildning och redoxreaktioner. Nedan följer en kortare genomgång av de olika typreaktionerna kopplat till de föreningar som förekommer i laborationen.

Typreaktioner

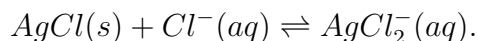
1. **Protolys** Vid en protolysreaktion sker en protonöverföring mellan en syra och en bas. Generellt tecknas en protolysreaktion som



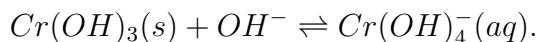
där syran HA och basen A^- utgör ett syra/baspar, tillika utgör syran H_3O^+ och basen H_2O ett annat syra/baspar. Alla kemiska föreningar kan betraktas som syror eller baser, eller mer exakt, anses ha syra- eller basegenskaper. Vissa föreningar kan reagera både som syra och

bas beroende på yttre omständigheter. De baser som används i laborationen är hydroxider, kromater (CrO_4^{2-}) och sulfider.

2. **Komplexbildning** Komplexbildning är vanlig vid reaktioner med övergångsmetaller. När kobolt löses upp i vatten bildas ett komplex, $Co^{2+}[H_2O]_6$. Detta faktum nämns vanligtvis inte på grund av bindningens svaga karaktär. En starkare bindning bildas då silverklorid utsätts för ett överskott av saltsyra,



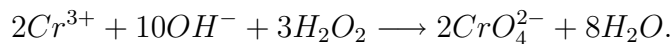
Vid överskott av saltsyra bildas ett silverkomplex, men när pH höjs så fälls silverkloriden åter ut. Ett annat exempel på komplexbildning är då krom(III)nitrat utsätts för en starkt basisk lösning,



I svagt basiska lösningar bildas en svårslöslig krom(III)lösning, $Cr(OH)_3$. Om pH är tillräckligt högt bildas en vattenlöslig förening, $Cr(OH)_4^-$.

3. **Redoxreaktion** En redoxreaktion består av en reduktion och en oxidation. Formellt sett består en redoxreaktion av en elektronöverföring mellan de olika typerna; således måste den totala laddningsändringen för alla ingående typer vara balanserad. Föreningen som donerar elektroner blir oxiderad och får en högre positiv laddning, medan föreningen som tar emot elektroner får en minskad laddning.

Om man har en starkt sur lösning ($pH \approx 14$) av kromjoner och tillsätter väteperoxid, H_2O_2 , sker en redox reaktion,



Krom(III) oxideras i exemplet ovan till krom(VI) och laddningsförändringen kompenseras genom sönderdelning av H_2O_2 till H_2O .

Löslighet av salter

För att finna vilka katjoner som ingår i ett okänt prov måste jonslagen separeras från varandra. Detta kan göras genom utfällningsreaktioner via komplexbildning eller redoxreaktioner. Företrädesvis används reaktioner som involverar utpräglade färgförändringar eller att det bildas gasformiga föreningar.

Det är en fördel att veta någonting om lösligheten hos föreningarna i de olika grupperna (se Fig B.1). I laborationsschemat används ett antal olika empiriska löslighetsregler. Emellertid är det ofta undantagen från dessa regler som används för att fälla ut önskat jonslag.

Empiriska löslighetsregler

Alla nitratföreningar är lösliga i vatten, något som gör dem till lämpliga utgångsmaterial. Generellt gäller följande:

- salter erhållna från starka syror (HCl , HNO_3 , H_2SO_4) är vanligen lösliga.
- neutrala salter erhållna från svaga syror (kolsyra, fosforsyra) är mer benägna att vara svårlösliga.
- alkalimetallsalter och ammoniumsalter tillikasa sura salter från diprotoniska syror (exempelvis sulfater) är generellt sett lösliga.

Table B.1: Löslighetsregler för oorganiska föreningar I, där *ss* står för svagt löslig [slightly soluble].

Lättlösliga föreningar	Exempel	Undantag
ammoniumföreningar	NH_4Cl	inga
klorider	$NaCl$	Ag^+ , Hg_2^{2+} , Pb^{2+} , $PbCl_2(ss)$
nitrater	$AgNO_3$	inga
sulfater	$PbSO_4$	Ca^{2+} , Sr^{2+} , Ba^{2+} , Pb^{2+} , Hg_2^{2+} , Ag^+ , $Ag_2SO_4(ss)$

Table B.2: Löslighetsregler för oorganiska föreningar II.

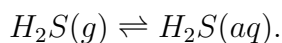
Svårösliga föreningar	Exempel	Undantag
karbonater, kromater, oxalater, fosfater	$PbCrO_4$	föreningar med grundämnen från Grupp 1 och NH_4^+
sulfider	CoS	föreningar med grundämnen från Grupp 1,2 och NH_4^+
hydroxider, oxider	$Cr(OH)_2$	föreningar med grundämnen från Grupp 1,2

Tabell B.1 och Tabell B.2 visar generella löslighetsregler i vatten vid $25^\circ C$ samt några undantag från dessa regler, hämtade från Atkins och Jones [5, p102].

Löslighet av sulfider kopplat till pH

En del sulfider är tillräckligt svårösliga så att även låga koncentrationer av S^{2-} ger utfällning. De sulfider som inte är så känsliga kan hållas från att fällas ut genom att noga avväga lösningens pH.

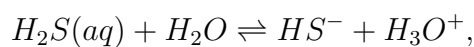
Under antagandet att källan till S^{2-} är H_2S sker följande reaktion när gas bubblas i lösningen:



Vidare, om lösningen har nått en mättnadspunkt, så sätts $[H_2S(aq)]$ till 0.1 M, oberoende av lösningens innehåll. Eftersom reaktionerna utförs vid rumstemperatur och atmosfärstryck (1 atm) fås följande jämviktskonstant,

$$K_f = \frac{[H_2S(aq)]}{P_{H_2S}} = 0.1 M \text{atm}^{-1}.$$

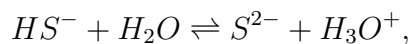
I lösningen sker följande reaktioner:



med jämviktskonstant,

$$K_{a1} = \frac{[H_3O^+][HS^-]}{[H_2S]} = 10^{-7} M$$

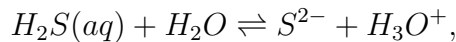
och



med jämviktskonstant,

$$K_{a2} = \frac{[H_3O^+][S^{2-}]}{[HS^-]} = 10^{-14} M.$$

Om de båda protolysreaktionerna ovan kombineras, blir den återstående reaktionen



och jämviktskonstanten ges av

$$K_{a1}K_{a2} = \frac{[H_3O^+][HS^-]}{[H_2S]} \frac{[H_3O^+][S^{2-}]}{[HS^-]} = \frac{[H_3O^+]^2[S^{2-}]}{[H_2S]},$$

vilket är ekvivalent med

$$[S^{2-}] = K_{a1}K_{a2} \frac{[H_2S]}{[H_3O^+]^2} = 10^{-7}10^{-14} \frac{0.1}{[H_3O^+]^2} = \frac{10^{-22}}{[H_3O^+]^2}.$$

Slutsatsen som kan dras är att ju lägre pH desto mindre är koncentrationen av S^{2-} och därmed också möjligheten till sulfidutfällning. Detta är vad separationsprocessen av sulfider i laborationen bygger på. De sulfider som ingår i laborationen och vars jämvikter påverkas nämnvärt av pH förändringar är MnS och PbS . Andra metaller, vars sulfider visar samma tendens är exempelvis Fe , Ni och Zn . I laborationen är CuS och PbS alltför svårslösliga för att påverkas av lösningens pH. Detta samband gäller även för sulfiderna av exempelvis Cd , Hg , As , Sn , Sb och Bi . Det finns mer att läsa om oorganisk kvalitativ analys angående sulfider och den intresserade läsaren hänvisas till Moeller *et al.* [18].

B.6 Reaktioner

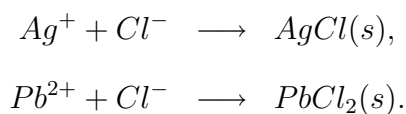
Laborationen är uppdelad i tre delar utifrån separationsschemat (Fig B.1). I den första laborationsdelen identifieras metalljoner som faller som klorider i sur miljö, i del 2 skapas sulfider i sur miljö och i den sista delen fälls metalljoner som hydroxider respektive sulfider i basisk miljö. Nedan beskrivs

de reaktioner som förekommer i de olika fallen. Alla utfällningsreaktioner som förekommer i laborationen har relativt höga löslighetskonstanter och beskrivs därmed nedan inte som jämvikter.

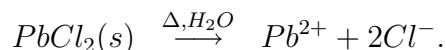
Laboration I: Silver- och blyföreningar

Silver(I)- och bly(II)joner faller ut som svårösliga klorider i sur miljö. Vid uppvärmning av kloriderna påverkas jämviktskonstanten för blyklorid betydligt, vilket leder till upplösning. Silverkloriden är däremot kvar oförändrad.

Då nitratlösningar av silver(I) och bly(II) surgörs med några droppar saltsyra bildas vita fällningar:



Blykloriden löses åter upp då den värms i kokande vattenbad,



På ovanstående sätt kan silver(I) och bly(II) separeras från varandra samt från övriga metalljoner i separationsschemat. Även om de båda kloridföreningarna är vita kan man i detta fall även se skillnader i deras utseende. Elever i utprovningen av laborationen³ observerade en skillnad i kornigheten hos föreningarna samt att det tog olika lång tid för utfällning att ske. Blykloridens utfällning beskrevs som snö, då de föll ut i storflockiga fingor. Några elever uppfattade även att bly(II) hade en söt doft.

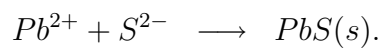
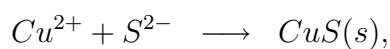
³Se Laborative chemistry in Natural Science



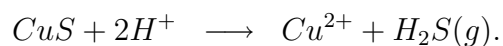
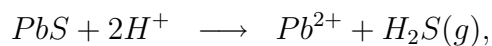
Figure B.2: Nyutfälld blyklorid.

Laboration II: Koppar- och blyföreningar

Koppar(II) och bly(II) faller båda som sulfider i sur miljö:



Bly(II) har en inte fullt så stor jämviktskonstant så risken finns, enligt Moeller *et al.* [18], att löst Pb^{2+} fortfarande finns i lösningen efter Laboration I. Kopparsulfid, CuS , är betydligt mer svårlöslig än bly sulfid, PbS ; bly sulfid kan lösas i 8M salpetersyra, medan det krävs koncentrerad svavelsyra för att lösa upp kopparsulfid:



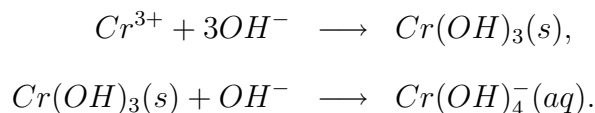
Upplöst Cu^{2+} ger vid komplexbindning med vatten upphov till en ljusblå lösning.

Laboration III: Krom- och koboltföreningar

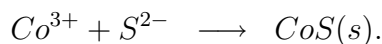
I denna laborationsdel används två buffertlösningar. Genom att blanda lika mängder 1M ammoniak och ammoniumklorid erhålls en buffert kring pH 9,



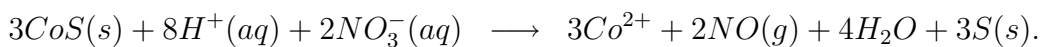
Om nitratlösningar av krom(III) och kobolt(II) buffras till pH 9 faller endast kromhydroxid ut. Ifall lösningen görs mer basisk bildas ett vattenlösligt kromhydroxokomplex:



Om den pH 9-buffrade lösningen av kobolt(II) låts reagera med sulfidjoner, fälls koboltsulfid ut,



För att skapa tillgång till sulfidjoner kan antingen svavelväte bubblas ner i lösningen (se avsnitt B.5) eller ammoniumsulfid, $(NH_4)_2S$, tillsättas. För att lösa upp sulfidfällningen tillsätts salpetersyra, vilket ger upphov till nitrösa gaser,



Krom(III) bildar inte någon sulfid utan på grund av den basiska miljön bildas den tidigare nämnda hydroxiden ovan, $Cr(OH)_3$. En buffertlösning kring pH 2 erhålls genom att blanda lika mängder 1M natriumvätesulfat och natriumsulfat. Genom att tillsätta denna buffertlösning till den utfällda hydroxiden frigörs åter igen krom(III) i lösningen,

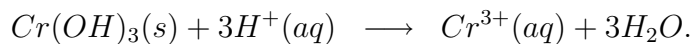
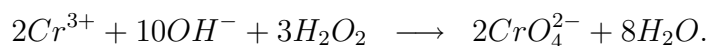




Figure B.3: Kromföreningar. Från vänster: Cr^{3+} löst i vatten; reaktionsblandning efter tillsats av H_2O_2 ; reaktionsblandning efter kokning (CrO_4^{2-}); $PbCrO_4$.

Genom att åter igen göra lösningen basisk och sedan tillsätta väteperoxid, oxideras krom(III) till krom(VI). Då bildas kromatjoner, CrO_4^{2-} , vilket ger en gulaktig lösning (Fig. B.3),



I sur miljö omvandlas kromat till dikromat, $Cr_2O_7^{2-}$, vilket ger en mer orange lösning och med bly(II) bildar en orange fällning. Bly(II) och kromat ger däremot en gul fällning (Fig. B.3),

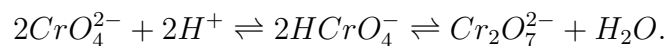




Figure B.4: Koboltföreningar. Från vänster: Co^{2+} löst i vatten; Co^{2+} ytterligare utspädd samt i mer basisk lösning; CoS ; delvis vattenupplöst Co^{2+} från CoS .

B.7 Materiel

För Laboration I behöver varje grupp:

- två vattenbägare (250 ml respektive 100 ml, till vattenbad och slask)
- en kokplatta
- ett provrörsställ med 2 centrifugrör
- en vattenflaska med destillerat vatten
- två pipetter (glas eller plast)
- två glasstavar
- totalt 3 droppflaskor á 20 ml med silvernitrat, blynitrat respektive saltsyra
- vattenställ för provrören eller en tång/värmehandske för att lyfta upp dem.

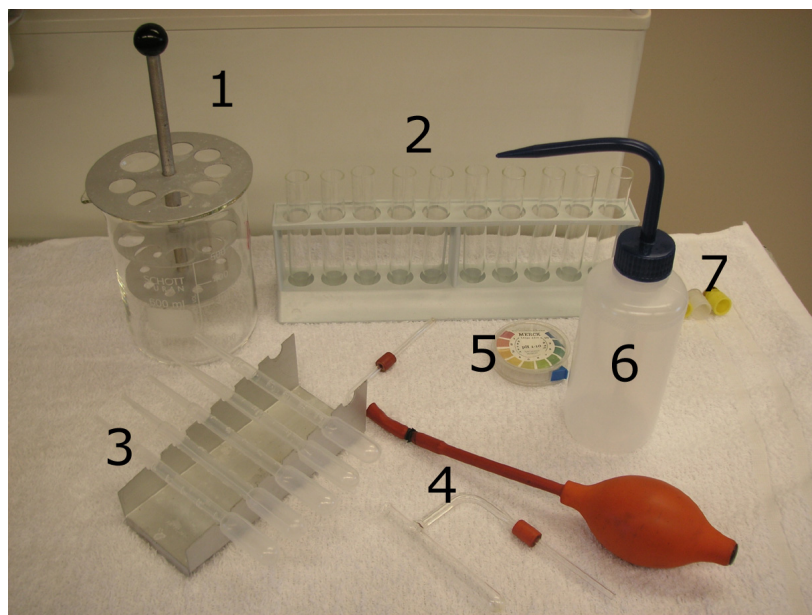


Figure B.5: Labutrustning. 1. Bägare med provrörshållare. 2. Provrörsställ med centrifugrör. 3. Pipetter i droppställ. 4. Gasledningsrör med pump. 5. pH band. 6. Dest. vattenflaska. 7. Provrörskorkar.

För Laboration II behöver varje grupp:

- två vattenbägare (250 ml respektive 100 ml, till vattenbad och slask)
- en kokplatta
- ett provrörsställ med 2 provrör och 2 centrifugrör
- en vattenflaska med destillerat vatten
- fyra pipetter (glas eller plast)
- två glasstavar
- pH-band/indikatorstickor (pH 1-14)

- totalt 2 droppflaskor á 20 ml med kopparnitrat respektive blynitrat.
- vattenställ för provrören eller en tång/värmehandske för att lyfta upp dem.

För Laboration III behöver varje grupp:

- två vattenbägare (250 ml respektive 100 ml, till vattenbad och slask)
- en kokplatta
- ett provrörsställ med 2 provrör och 2 centrifugrör
- en vattenflaska med destillerat vatten
- fyra pipetter (glas eller plast)
- två glasstavar
- pH-band/indikatorstickor (pH 1-14)
- totalt 3 droppflaskor á 20 ml med kromnitrat respektive koboltnitrat samt natriumhydroxid.
- vattenställ för provrören eller en tång/värmehandske för att lyfta upp dem.

Droppflaskorna behöver inte rymma mer än 20 ml lösning. Vid upphettning på vattenbad kan man även använda ett provrörsställ (se Fig. B.5). Ytterligare förslag på labutrustning finns i Fig. B.5. Till skapandet av svavelväte används förslagsvis en gasledningsflaska, där reaktionen mellan järnsulfid och saltsyra låts ske. Sätt en slang i var ände av flaskan. Sätt

sedan en glaspipett ytterst i en av slangarna. Pipetten används till att bubbla ner gasen i provlösningarna. Glöm inte att hålla för den andra slangen vid gasöverföringen. Dessutom behövs det tillgång till en provrörscentrifug.

B.8 Förberedelser

Laboration I

- Blanda 0.1 M lösningar av silvernitrat och blynitrat. Tänk på att arbeta i dragskåp eller under punktsug eftersom det är ohälsosamt att inandas damm från dessa föreningar. Vid laborationen används bara några ml av respektive nitratlösning per grupp.
- Blanda saltsyra och häll i droppflaskor.
- Volymen saltsyra kan variera beroende på koncentration. Ifall 5M saltsyra används, behövs bara någon droppe per grupp. Ifall exempelvis 0.5 M saltsyra används krävs 2-3 ml. Blanda i såfall 20 ml 0.5 M lösning per grupp. Det enklaste är att blanda en liter av var lösning och fördela dessa i droppflaskor. Så har du till egen testning också.
- Fyll droppflaskor (ca 20 ml) med lösningar.
- Ställ fram på ett rullbord samtlig utrustning (se avsnitt B.7).

Laboration II

- Blanda 0.1M lösningar av kopparnitrat och blytnitrat. Tänk på att arbeta i dragskåp eller under punktsug eftersom det är ohälsosamt att inandas damm från dessa föreningar. Vid laborationen används bara några ml av respektive nitratlösning per grupp.
- Blanda 8M salpetersyra och förvara på lämplig plats.

- Blanda 1M natriumsulfat- respektive 1M natriumvätesulfatlösning, var grupp behöver några ml.
- Fyll droppflaskor (ca 20 ml) med lösningar.
- Ställ fram på ett rullbord samtlig utrustning (se avsnitt B.7).
- Placera de starka syrorna på lämplig plats.

Laboration III

- Blanda 0.1M lösningar av kromnitrat och koboltnitrat. Tänk på att arbeta i dragskåp eller under punktsug eftersom det är ohälsosamt att inandas damm från dessa föreningar. Vid laborationen används bara några ml av respektive nitratlösning per grupp.
- Blanda 5M natriumhydroxid och häll i droppflaskor.
- Blanda 1M natriumsulfat- respektive 1M natriumvätesulfatlösning, var grupp behöver några ml.
- Blanda 1M ammoniak- respektive ammoniumkloridlösning, var grupp behöver några ml.
- Fyll droppflaskor (ca 20 ml) med lösningar.
- Ställ fram på ett rullbord samtlig utrustning (se avsnitt B.7).
- Placera väteperoxiden samt den koncentrerade syran på lämplig plats.

Table B.3: Kemikalier före reaktion. (Laboration I)

Namn	Kemisk formel	Konc.	Klassificering
Silvernitratt	$AgNO_3$	15–10%	R 36/38 (50) S (1/2) 25 46
Bly(II)nitrat	$Pb(NO_3)_2$	5–0.5%	Giftig R 61, R 33 (50) S 53, 45
Saltsyra	HCl	25–10%	R 36/37/38 S (1/2) 9, 26, 36/37/39, 45

B.9 Riskanalys för Laboration I

Risk- och säkerhetsklassificering

I laborationen används endast 0.1 M nitratlösningar samt 5M saltsyra. Klassificeringen av föreningarnas risker vid utförandet och säkerhetsbeskrivning är gjord i förhållande till föreningarnas koncentration. Datan i Tabell B.3 och B.4 är hämtade från *Kemiska Ämnen 4.0*, en databas skapad av Arbetskyddsmyndigheten.

Bedömning och skyddsåtgärder

Utrustning

Kokplatta med vattenbad. Förvara inte kemikalielösningar för nära kokplattan.

Vid utförandet

Table B.4: Kemikalier efter reaktion. (Laboration I)

Namn	Kemisk formel	Konc.	Klassificering
Silverklorid	$AgCl$	100–0%	inga risker R 50
Bly(II)klorid	$PbCl_2$	5–0.5%	Giftig R 61, R 33 (50) S 53, 45

Använd laborationsrock eller förkläde, skyddsglasögon. Undvik (som alltid) hudkontakt med kemikalielösningarna. Silverniträt ger svarta fläckar på huden. Laborationen utförs i liten skala (1 ml lösning), vilket gör att hanteringen av kemikalierna minimeras. Lösningarna som används är färdigblandade inför laborationen och hällda i droppflaskor. Att tänka på under laborationens:

- Förberedelse: hämta färdigblandade lösningar och utrustning under ordnade former, inget spring.
- Genomförande: ha måttlig uppvärmning av vattenbadet. Vid stormkokning kan vatten stänka och orsaka skollskador. Även kan provrör börja hoppa och gå sönder eller innehållet hällas ut i badet. Vid tvättning av fällningar, samlas tvättvätskan upp i en bägare för senare destruering.
- Avslutning: Stäng av kokplattan, låt vattnet svalna. Ställ tillbaka använda droppflaskor och utrustning på avsedd plats.

Avfallshantering och diskning

Lösningar av bly- eller silverföreningar hälls i destrueringsbägaren. Rester i provrör sköljs ut med så liten mängd vatten som möjligt och hälls i samma bägare. Sedan rengörs provrören med diskmedel. Slaskbägarens innehåll destrueras på angiven plats.

Sammanfattande bedömning

Laborationen bedöms som mindre riskfylld i och med de låga koncentrationerna på kemikalierna.

Riskbedömning utförd av

Namn: _____

Skola: _____

Datum: _____

Table B.5: Kemikalier före reaktion. (Laboration II)

Namn	Kemisk formel	Konc.	Klassificering
Koppar(II)nitrat	$Cu(NO_3)_2$	25–10%	R 322, R (50) S 2
Bly(II)nitrat	$Pb(NO_3)_2$	5–0.5%	R 61, R 33 (50) S 53,45
Järn(II)sulfid	FeS	100–0%	ofarlig
Salpetersyra	HNO_3	5–1%	R 36/38 S (1/2) 23 26 36 45
Svavelsyra	H_2SO_4	15%	R 34 R 14 S (1/2) 26, 45, 30

B.10 Riskanalys för Laboration II

Risk- och säkerhetsklassificering

I laborationen används endast 0.1 M nitratlösningar samt 8 M salpetersyra. Svavelsyran är dock högkoncentrerad. Klassificeringen av föreningarnas risker vid utförandet och säkerhetsbeskrivning är gjord i förhållande till föreningarnas koncentration. Datan i Tabell B.5 och B.6 är hämtade från *Kemiska Ämnen 4.0*, en databas skapad av Arbetarskyddsnämnden.

Laborationen innehåller upplösning av sulfider vid bildande av svavelväte. Därmed måste laborationen utföras i dragskåp eller under punktsug.

Table B.6: Kemikalier efter reaktion. (Laboration II)

Namn	Kemisk formel	Konc.	Klassificering
Koppar(II)sulfid	CuS	100–0%	ingenting
Bly(II)sulfid*	PbS	100%	R 20/21/22, 40 S 36, 22, 7, 26
Svavelväte	H_2S	10–5%	R 23, R12, 50 S (1/2), 9, 16, 28, 36/37, S 45, 61
Blyulfat	$PbSO_4$	5–0.5%	R 61, R 30, (50) S 53, 45

*Data är hämtade från Aylward och Findlays *SI Chemical Data*, då ingen information kunde hittas i Arbetarskyddsnämndens databas *Kemiska Ämnen 4.0*.

Bedömning och skyddsåtgärder

Utrustning

Kokplatta med vattenbad. Förvara inte kemikalielösningar för nära kokplattan.

Vid utförandet

Använd laborationsrock eller förkläde, skyddsglasögon. Undvik (som alltid) hudkontakt med kemikalielösningarna. Starka syror förvaras på avsedd plats och används under uppsikt av lärare. Laborationen utförs i liten skala (1 ml lösning), vilket gör att hanteringen av kemikalierna minimeras. Lösningarna som används är färdigblandade inför laborationen och hällas i droppflaskor. Att tänka på under laborationens:

- Förberedelse: hämta färdigblandade lösningar och utrustning under ordnade former, inget spring.
- Genomförande: korka igen alla centrifugrör med sulfidfällningar då de förflyttas mellan dragskåp. Ställ centrifugen i ett dragskåp, som ligger nära till hands för alla laboranter. Ha måttlig uppvärmning av vattenbadet. Vid stormkokning kan vatten stänka och orsaka skollskador. även kan provrör börja hoppa och gå sönder eller innehållet hållas ut i badet. Vid tvättning av fällningar, samlas tvättvätskan upp i en bägare för senare destruering.
- Avslutning: Stäng av kokplattan, låt vattnet svalna. Ställ tillbaka använda droppflaskor och utrustning på avsedd plats.

Avfallshantering och diskning

Lösningar av koppar- eller blyföreningar hålls i destrueringbägaren. Rester i provrör sköljs ut med så liten mängd vatten som möjligt och hålls i samma bägare. Sedan rengörs provrören med diskmedel. Slaskbägarens innehåll destrueras på angiven plats. Överblivna syralösningar neutraliseras med en godtycklig bas (natriumhydroxid, ammoniak) innan de hålls ut i avloppet. Överbliven sulfidfällning löses i syra innan vidare destruering. Detta steg bör lättast göras av lärare.

Sammanfattande bedömning

Laborationen bedöms som måttligt riskfylld på grund av användandet av starka syror. Risker minimeras då läraren utför tillsatser och hantering av koncentrerade syror. Den koncentrerade syran används endast i extrauppgiften, lagd sist i laborationen.

Riskbedömning utförd av

Namn:_____

Skola:_____

Datum:_____

B.11 Riskanalys för Laboration III

Risk- och säkerhetsklassificering

I laborationen används endast 0.1 M nitratlösningar. Ammoniaklösningen är 5M, och natriumvätesulfat- respektive natriumsulfatlösningen är 1M. Överbliven väteperoxid kokas bort till syrgas och vattenånga. Klassificeringen av föreningarnas risker vid utförandet och säkerhetsbeskrivning är gjord i förhållande till föreningarnas koncentration. Datan i Tabell B.7 och B.8 är hämtade från *Kemiska Ämnen 4.0*, en databas skapad av Arbetarskyddsmyndigheten.

Bedömning och skyddsåtgärder

Utrustning

Kokplatta med vattenbad. Förvara inte kemikalielösningar för nära kokplattan.

Vid utförandet

Använd laborationsrock eller förkläde, skyddsglasögon. Undvik (som alltid) hudkontakt med kemikalielösningarna. Starka syror förvaras på avsedd plats och används under uppsikt av lärare. Laborationen utförs i liten skala (1 ml lösning), vilket gör att hanteringen av kemikalierna minimeras. Lösningarna som används är färdigblandade inför laborationen och hällda i droppflaskor. Att tänka på under laborationens:

- Förberedelse: hämta färdigblandade lösningar och utrustning under ordnade former, inget spring.
- Genomförande: korka igen alla centrifugrör med sulfidfällningar då de förflyttas mellan dragskåp. Ställ centrifugen i ett dragskåp, som lig-

Table B.7: Kemikalier före reaktion. (Laboration III)

Namn	Kemisk formel	Konc.	Klassificering
Krom(III)nitrat	$Cr(NO_3)_3$	100–25%	R 22, R 8 S 20
Kobolt(II)nitrat	$Co(NO_3)_2$	10–1%	R 43, R 8 S 45
Järn(II)sulfid	FeS	100–0%	ingenting
Väteperoxid	H_2O_2	60–20%	R 34 S (1/2) 28, 36/39, 45, 3
Natriumvätesulfat	$NaHSO_4$	10–5%	R 36/37/38 S (2) 22, 24, 37
Natriumsulfat	$NaSO_4$	100–0%	ingenting
Ammoniak	NH_3	10–5%	R 36/37/38 R 50 S (1/2) 26, 36/37/39, 45, S 61, 7
Natriumhydroxid	$NaOH$	50%	R 34 S 26, 27, 36/37/39

Table B.8: Kemikalier efter reaktion. (Laboration III)

Namn	Kemisk formel	Konc.	Klassificering
Krom(III)hydroxid	$Cr(OH)_3$	100–0%	ingenting
Bly(II)kromat	PbS	1–0.5%	R 340, R 50, 53 S 53, 45, 60, 61
Kobolt(II)sulfid	CoS	100–1%	R 43 S (2), 24, 37
Svavelväte	H_2S	10–5%	R 23, R12, 50 S (1/2), 9, 16, 28, 36/37, S 45, 61

ger nära till hands för alla laboranter. Ha måttlig uppvärmning av vattenbadet. Vid stormkokning kan vatten stänka och orsaka skollskador. även kan provrör börja hoppa och gå sönder eller innehållet hållas ut i badet. Vid tvättning av fällningar, samlas tvättvätskan upp i en bägare för senare destruering. Vidta försiktighet vid tillsatsen av väteperoxid. Tillsätt droppvis, då det lätt börjar bubbla i provröret.

- Avslutning: Stäng av kokplattan, låt vattnet svalna. Ställ tillbaka använda droppflaskor och utrustning på avsedd plats.

Avfallshantering och diskning

Lösningar av krom- eller koboltföreningar hålls i destrueringsbägaren. Rester i provrör sköljs ut med så liten mängd vatten som möjligt och hålls i samma bägare. Sedan rengörs provrören med diskmedel. Slaskbägarens innehåll destrueras på angiven plats. Överblivna syralösningar neutraliseras med en

godtycklig bas (natriumhydroxid, ammoniak) innan de hälls ut i avloppet. Överbliven sulfidfällning löses i syra innan vidare destruering. Detta steg bör lättast göras av lärare.

Sammanfattande bedömning

Laborationen bedöms som måttligt riskfylld på grund av användandet av starka syror. Risker minimeras då läraren utför tillsatser och hantering av koncentrerade syror. Den koncentrerade syran används endast i sista steget i laborationen.

Riskbedömning utförd av

Namn: _____

Skola: _____

Datum: _____

Table B.9: Kostnadsanalys

Namn	Casnr	Mängd	Pris	Konc.
Ammoniak	7664-41-7	100 ml	356	2,0 M
Blynitrat	10099-74-8	5 g	98	99%
Järnsulfid	1317-37-9	250 g	134	+40 mesh
Koboltnitrat hexahydrat	10026-22-9	100 g	313	98%
Kopparnitrat hydrat	—	5 g	177	99.999%
Kromnitrat nonahydrat	7789-02-8	100 g	115	99%
Natriumhydroxid	1310-73-2	500 g	125	97%
Natriumsulfat decahydrat	7727-73-3	500 g	144	97%
Natriumvätesulfat	7681-38-1	500 g	125	Tech
Salpetersyra	7697-37-2	25 ml	319	90%
Saltsyra	7647-01-0	500 ml	276	37%
Silvernitratt	7761-88-8	500 ml	246	0,1 M
Svavelsyra	7664-93-9	1 l	352	95–98%
Väteperoxid	7722-84-1	100 ml	143	30 wt%

B.12 Kostnadsanalys

I Tabell B.9 ges en översiktlig bild över kostnaderna för respektive kemikalie i laborationen. I vissa lägen skilde det enbart några få kronor mellan exempelvis 5g och 250g. I dessa fall återges kostnaden för den större mängden. Priserna är hämtade från företaget Aldrich (1999-2000).

B.13 Referenser

Arbetarskyddsnämnden, *Kemiska Ämnen 4.0*; en databas över kemiska föreningar inklusive kemiska och fysikaliska data samt risker och hantering.

Aylward och Finlay. (2000) *SI Chemical Data*, 4th edition.

Appendix C

Questionnaire

In the beginning of the chemistry section in Science Studies A, the students were given a questionnaire (Fig. C) where they were to measure their own knowledge of different chemistry sections that are included in their course. They (the students) were to evaluate their knowledge in chemistry in average and then on concepts like the periodic system, metals and non-metals, atoms and molecules, ions and chemical bonding. There were questions about their interest in chemistry and on how they want to work during chemistry lessons. This form was created together with the teacher who were interested in these aspects of her teaching. The results of the questionnaire are presented in Table C.1. After the laboratory testing the students once again answered the questions about their knowledge in chemistry (Table C.2).

Namn: _____

Naturkunskap A

Första kapitlet handlar som du ser om kemi. Innan vi börjar med den oorganiska kemin vill jag gärna veta vad du kan i kemi, om du är intresserad av kemi, hur du vill arbeta och vilka mål du har med dina studier i kemi. Sätt en ring runt det svar som passar dig bäst.

Kunskaper

- | | | | | | |
|----|--|-------|-----|------------|----------|
| 1. | Mina kunskaper i kemi är | svaga | bra | mycket bra | utmärkta |
| 2. | Mina kunskaper om periodiska systemet är | svaga | bra | mycket bra | utmärkta |
| 3. | Mina kunskaper om metaller och ickemetaller är | svaga | bra | mycket bra | utmärkta |
| 4. | Mina kunskaper om atomer och molekyler är | svaga | bra | mycket bra | utmärkta |
| 5. | Mina kunskaper om joner är | svaga | bra | mycket bra | utmärkta |
| 6. | Mina kunskaper om kemisk bindning är | svaga | bra | mycket bra | utmärkta |
| 7. | Mina kunskaper om syror och baser är | svaga | bra | mycket bra | utmärkta |

Intresse mm

- | | | | | | |
|-----|---------------------------|-----------|-------------|------------|--------------------|
| 8. | Jag tycker kemi är | tråkigt | sådär | ganska kul | roligt |
| 9. | Jag tycker kemi är | lätt | lagom svårt | svårt | alldeles för svårt |
| 10. | Jag vill lära mer om kemi | inte alls | ok | gärna | mycket gärna |

Arbetssätt

- | | | | | | |
|-----|---|-----------|--------|---------|--------------|
| 11. | Jag vill arbeta enbart med frågorna i boken | inte alls | ok | gärna | mycket gärna |
| 12. | Jag vill laborera | inte alls | ibland | ofta | varje gång |
| 13. | Jag vill ha skriftligt prov i kemi | inte alls | gärna | absolut | |
| 14. | Jag vill göra ett eget arbete i kemi | inte alls | gärna | absolut | |

NN-06

Table C.1: The students' opinions on their knowledge in different chemistry sections, their interest in chemistry and the way of working during the lessons prior to the testing of the laboratory work. The students' answers are denoted A, B, C resp. D and are referring to the student questionnaire, Fig. C, alternatives (from the left) and given in percentages.

<i>Kunskaper</i>	A	B	C	D	Totalt
Mina kunskaper i kemi är	38.4	38.4	23.1	0.0	100.0
Mina kunskaper om periodiska systemet är	69.2	15.4	15.4	0.0	100.0
Mina kunskaper om metaller och ickemetaller är	23.1	61.5	15.4	0.0	100.0
Mina kunskaper om atomer och molekyler är	53.8	38.4	7.7	0.0	100.0
Mina kunskaper om joner är	61.5	38.4	0.0	0.0	100.0
Mina kunskaper om kemisk bindning är	76.9	23.1	0.0	0.0	100.0
Mina kunskaper om syror och baser är	46.2	38.4	15.4	0.0	100.0
<i>Intresse</i>	A	B	C	D	Totalt
Jag tycker kemi är	15.4	46.2	38.4	0.0	100.0
Jag tycker kemi är	0.0	53.8	46.2	0.0	100.0
Jag vill lära mer om kemi	7.7	53.8	30.8	7.7	100.0
<i>Arbetsätt</i>	A	B	C	D	Totalt
Jag vill enbart arbeta själv med frågorna i boken	61.5	30.8	0.0	7.7	100.0
Jag vill laborera	0.0	23.1	46.2	30.8	100.0
Jag vill ha skriftligt prov i kemi	76.9	23.1	0.0	-.	100.0
Jag vill göra ett eget arbete i kemi	76.9	23.1	0.0	-.	100.0

Table C.2: The students' opinions on their knowledge in different chemistry sections and in chemistry in general after the testing of the laboratory work. The students' answers are denoted A, B, C resp. D and are referring to the student questionnaire, Fig. C, alternatives (from the left) and given in percentages.

<i>Kunskaper</i>	A	B	C	D	Totalt
Mina kunskaper i kemi är	11.1	55.6	33.3	0.0	100.0
Mina kunskaper om periodiska systemet är	22.2	55.6	22.2	0.0	100.0
Mina kunskaper om metaller och ickemetaller är	0.0	55.6	44.4	0.0	100.0
Mina kunskaper om atomer och molekyler är	0.0	44.4	55.6	0.0	100.0
Mina kunskaper om joner är	0.0	77.8	22.2	0.0	100.0
Mina kunskaper om kemisk bindning är	0.0	77.8	22.2	0.0	100.0
Mina kunskaper om syror och baser är	11.1	55.6	33.3	0.0	100.0

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